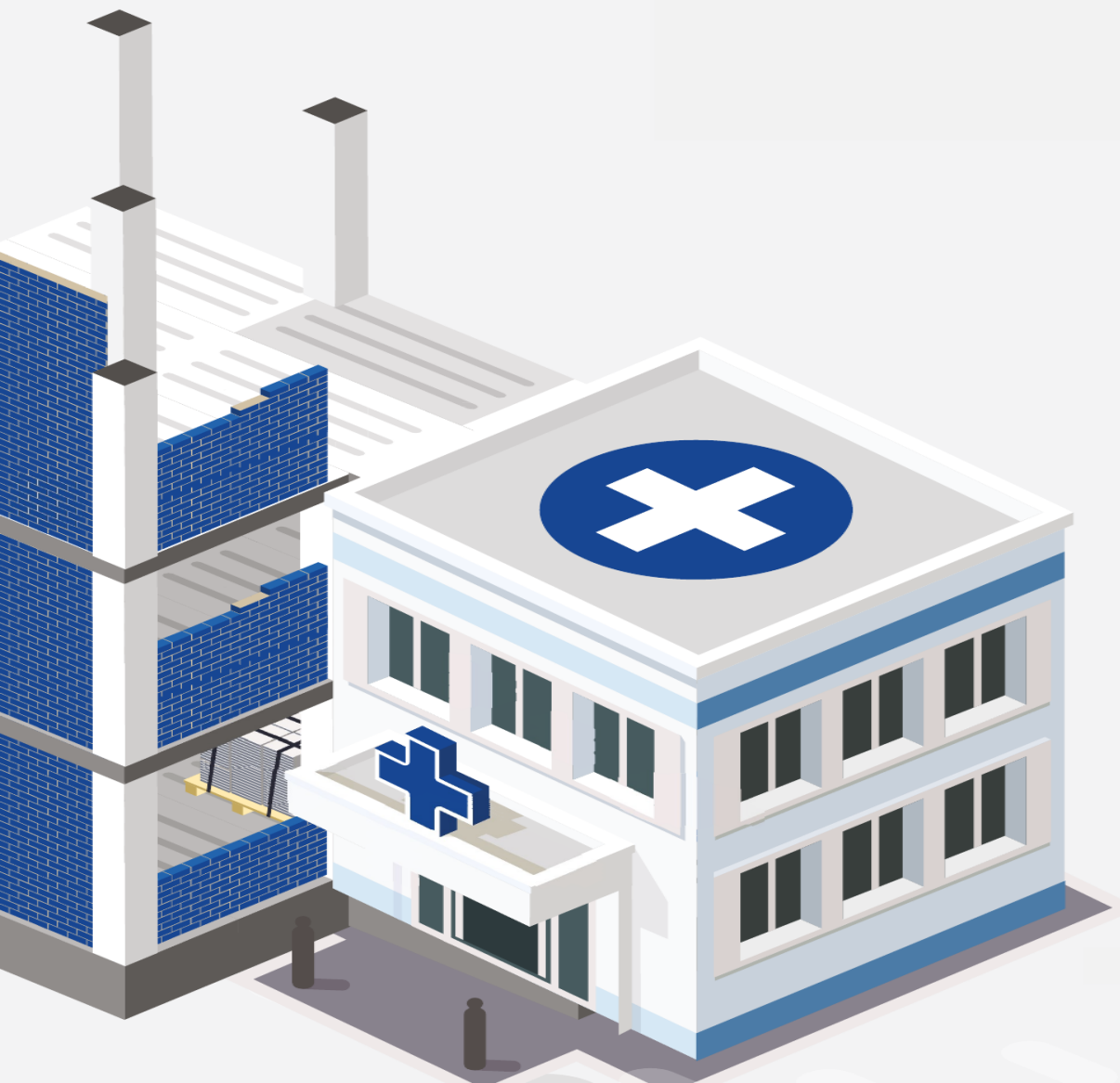




NARAYANA HRUDAYALAYA LTD.

Business Analysis & Valuation Report

Introduction

Narayana Hrudayalaya Limited together with its subsidiaries and associates is primarily engaged in business of rendering medical and healthcare services. Narayana Hrudayalaya Limited, the flagship company of the Group, was incorporated on 19th July 2000 by Dr. Devi Prasad Shetty as its Founder. It has a network of 23 multi-specialty and super specialty hospitals with 7 heart centers spread across 18 locations in India and 1 multi-specialty hospital in Cayman Islands.

The organization aims to provide comprehensive healthcare services underpinned by a mission centered on accessibility and quality of care. Narayana Hrudayalaya has a strong geographical presence in India. Their recent operational highlights include successful implementation of robotic surgeries and advanced treatment techniques, reflecting their commitment to clinical excellence.

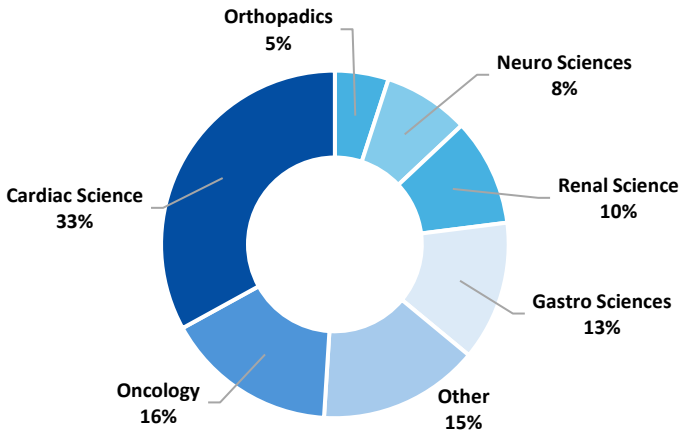
Business Segment

Narayana Hrudayalaya Limited include advanced surgical procedures and treatments in various specialties such as cardiac sciences, oncology, and orthopedics. (NHL) operates 18 owned hospitals, 7 heart centers, and 17 clinics across India, with major facilities in the South (5 hospitals, 3 heart centers, and 11 clinics), East (7 hospitals, 4 clinics, 1 dialysis center), North (4 hospitals, 1 clinic), and West (2 hospitals), totaling 6,254 beds. Internationally, it has a multi-specialty hospital in the Cayman Islands with 110 beds. With a team of 4,138 doctors.

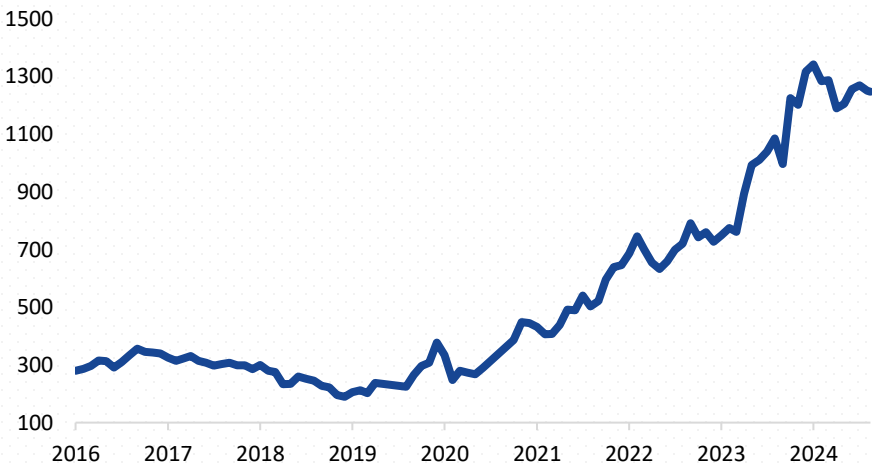
NHL is dedicated to delivering accessible healthcare. The revenue mix is 70% from owned hospitals, 27% from operated hospitals, 2% from heart centers, and 1% from other sources, showcasing a focus on core facilities while expanding through specialty centers and international reach.

Source: Trendline, go India

Therapeutic Area Wise Break-up Domestic



Stock Performance



Source: Tijori, yahoo finance

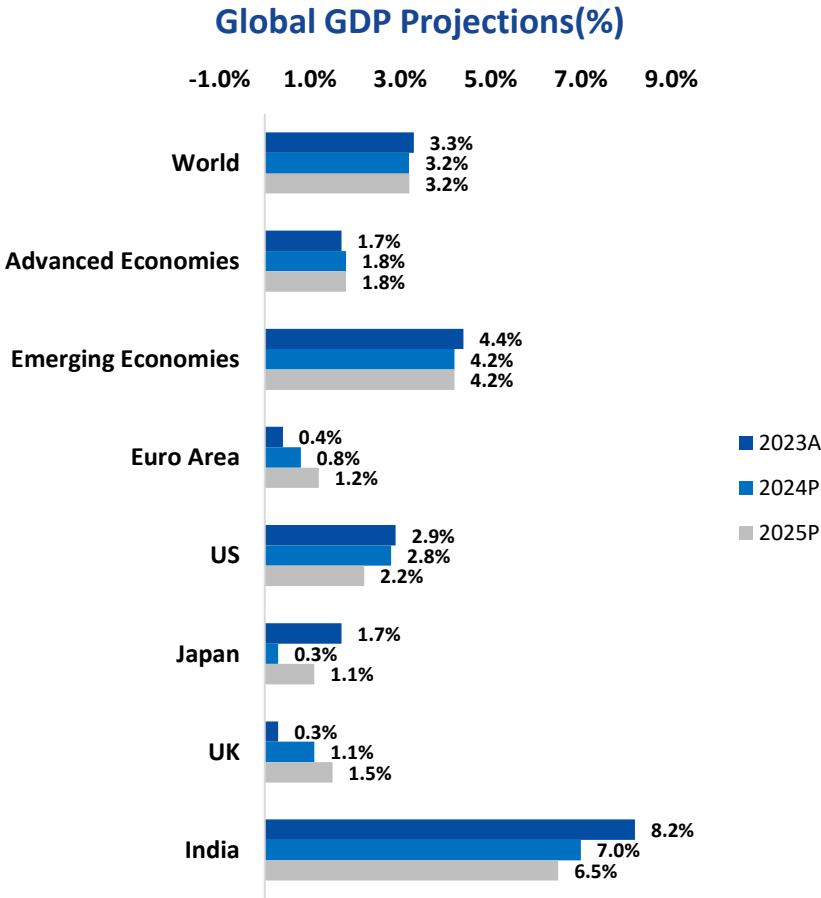
Overview

Global growth is expected to remain stable at 3.2% in 2024 and 2025. U.S. growth has improved, reflecting stronger performance in advanced economies, though the outlook for major European countries has been downgraded. Emerging markets and developing economies are facing challenges due to disruptions in commodity production (especially oil), shipping delays, conflicts, civil unrest, and extreme weather. Growth projections for regions like the Middle East, Central Asia, and Sub-Saharan Africa have been revised downward by the IMF. This uneven growth landscape indicates that the global economy is experiencing external shocks and heightened geopolitical uncertainties, with continued impacts from supply chain disruptions and climate change.

These downgrades are partially offset by an upgraded forecast for emerging Asia, driven by high demand for semiconductors and electronics, fueled by significant investments in AI. However, many economies are facing structural issues, including aging populations and weak productivity, which may slow growth. Cyclical imbalances are decreasing this year, which is beneficial for various countries, as inflation rates are converging globally. While goods prices are stable, key regional service price inflation remains high, highlighting sectoral dynamics that call for corresponding adjustments in monetary policy. Risks to the global economy are skewed to the downside due to high policy uncertainty.

Ongoing geopolitical tensions could disrupt disinflation, while a deeper-than-expected contraction in China's property sector may hurt consumer sentiment and have negative global repercussions, given China's significant role in global trade. Additionally, rising social tensions may affect confidence and delay necessary structural reforms. Many countries require fiscal policy adjustments to maintain debt sustainability and rebuild fiscal buffers. Global cooperation is essential to accelerate the green transition, support debt restructuring, and mitigate risks from geoeconomic fragmentation to ensure inclusive growth benefits, according to the IMF.

Source: IMF WEO, Company Analysis



Source: IMF WEO

Overview

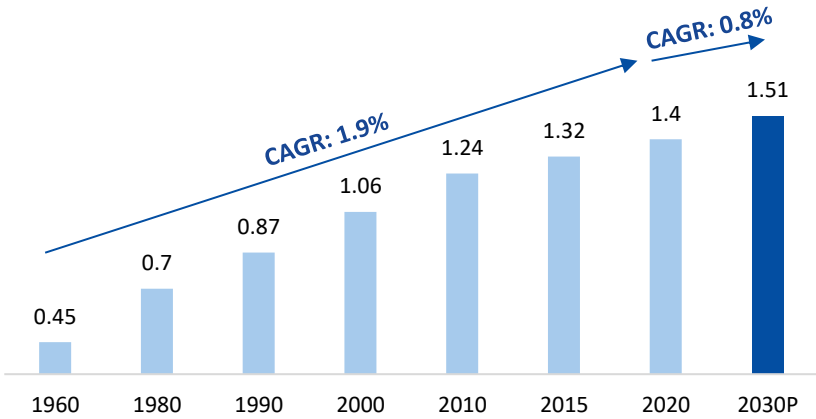
The Indian economy is performing better than expected, with a growth rate of 7.0% in this fiscal year. However, real GDP growth is projected to moderate to 6.5% in fiscal 2025. This projection is supported by increasing consumer spending, especially in rural areas, as inflation subsides and agricultural output improves due to favorable monsoon conditions. The Indian economy relies heavily on consumption, with about 60% of GDP coming from spending on goods and services. Currently, rural growth is outpacing urban growth.

Despite the global economic challenges viz. Uncertainty from adverse geopolitical developments. GDP growth is expected to reach 6.5% in FY25, as per IMF GDP forecast. Uneven economic growth for key trade partners and an escalation of the ongoing Red Sea crisis can be a drag on exports. with a rising share of salaried positions and of services subsectors demanding higher qualifications, including business and professional services in the areas of technology and finance. Female participation in the labor force has also increased, particularly in rural areas.

The Indian economy benefits from robust policies implemented by the Reserve Bank of India (RBI), which plays a crucial role in maintaining stability through its adept monetary policy framework. Following the Federal Reserve's rate cut, India kept its rates unchanged, with the RBI adjusting rates based on inflation trends. The RBI aims to achieve its medium-term inflation target of 4% (Consumer Price Index) within a tolerance band of +/- 2%, while also supporting growth. Meanwhile, the central bank has left its inflation forecast for this fiscal year unchanged at 4.5%, despite concerns over rising food prices and intensifying geopolitical tensions that could disrupt energy supplies and push crude oil prices higher.

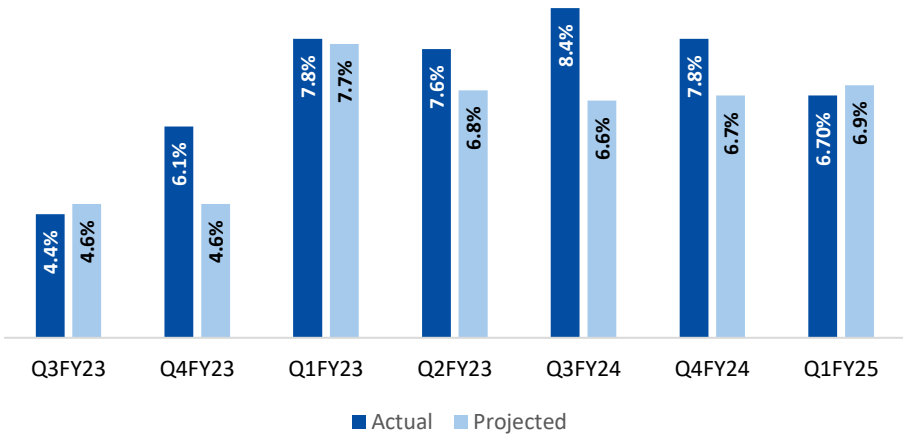
Source: IMF WEO, Deloitte, Crisil, IBEF

India's Population, (Billion)



Source: Crisil

India GDP Quarterly Growth - Actual vs Projected



Source: Investing.com

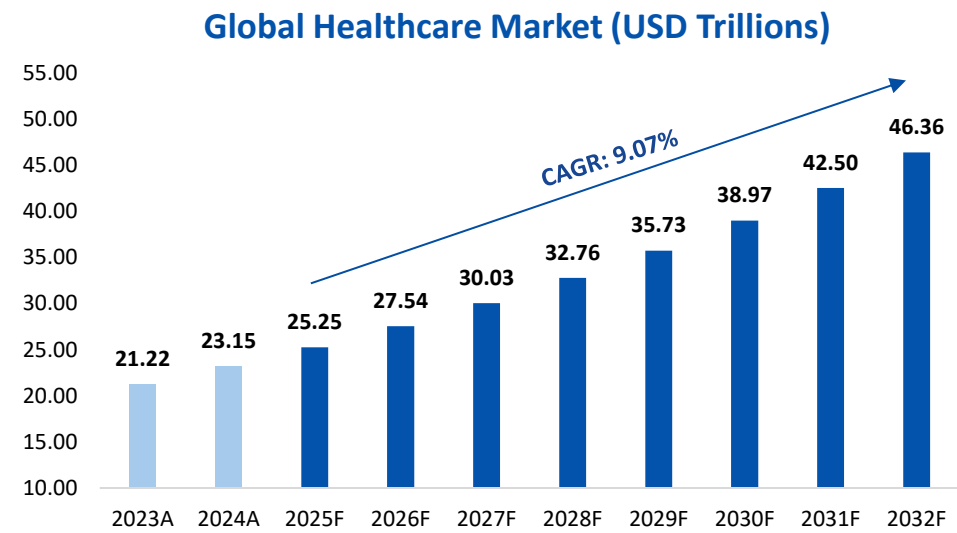
Healthcare Sector - Global

Overview

The Global Healthcare Market Size was valued at USD 21.222.5 Trillion in 2023, and is expected to reach USD 44.760.73 Trillion by 2032, and grow at a CAGR of 9.07% over the forecast period 2024-2032.

The healthcare industry is one of the fastest growing industries globally, driven by advancements in medical technology, growing demand for quality healthcare services, and increasing global population.

North America is the largest healthcare market, mainly because the U.S. spends heavily on healthcare. Europe follows, with many countries offering government-funded care. The Asia-Pacific region, especially China and India, is growing the fastest as incomes rise and more money is invested in healthcare. Though challenges like high costs, regulations, and uneven access remain, the global healthcare market keeps growing, supported by new technology and improved access to care.



Source: S&S Insiders

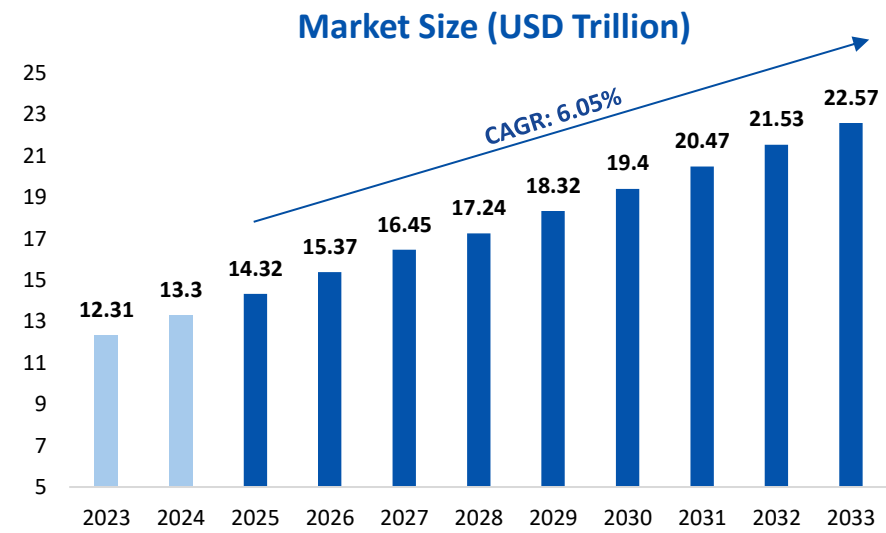
Hospital Industry - Global

Overview

The Global hospital services market was valued at USD 12.31 trillion in 2023 and is expected to reach around USD 22.57 trillion by 2033 and is poised to grow at a compound annual growth rate (CAGR) of 6.05% during the forecast period 2024 to 2033.

The global hospital services market is growing due to factors like the rise in cancer cases, more knee replacement surgeries, and the availability of advanced medical devices. As healthcare becomes more affordable and people become more aware, demand for medical services is increasing, boosting the economy in many countries.

Hospitals play a central role in this market, attracting a large consumer base and driving revenue within the healthcare industry. Overall, hospitals are a critical part of the healthcare industry, influencing not only patient care but also related sectors like pharmaceuticals, medical devices, and healthcare management services.



Source: Precedence Research

Overview

India’s healthcare industry, which includes hospitals, medical devices, health insurance, clinical trials, telemedicine, and medical tourism, has grown rapidly over the past five years. Today, healthcare is one of the largest sectors in India, both in revenue and employment. By FY 2023, the Indian healthcare market was valued at around USD 216 billion, and it is projected to reach approximately USD 456 billion by 2028, growing at an annual rate of 16.12%.

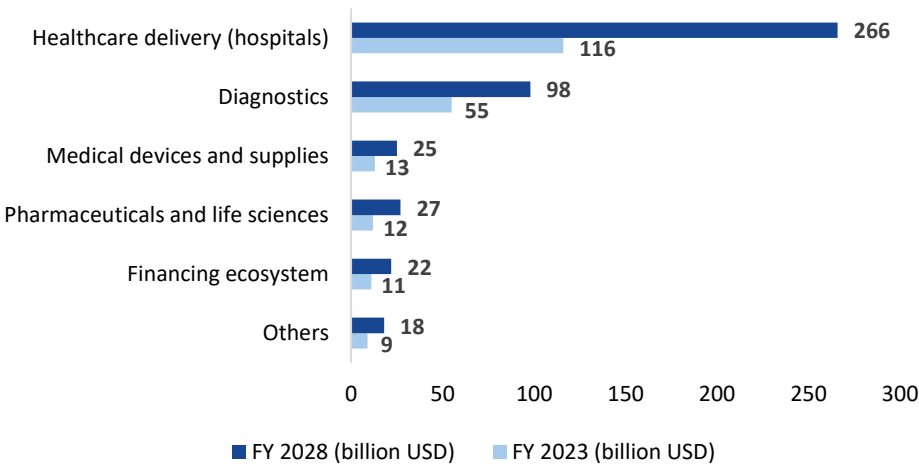
This growth is mainly due to the rising demand for better-quality and specialized healthcare services. The need for healthcare professionals in India is expected to double by 2030, both in India and abroad, due to a shortage of healthcare workers. Currently, India has only 1.7 nurses per 1,000 people and a doctor-to-patient ratio of 1:1,500. Key factors driving growth in the healthcare sector include an aging population, a growing middle class, the increase in lifestyle-related diseases, more public-private partnerships, and the rapid adoption of digital healthcare technologies like telemedicine. There has also been an increase in investor interest and foreign direct investment (FDI) in healthcare over the past 20 years.

Additionally, COVID-19 has changed how people view personal health, hygiene, health insurance, fitness, and regular medical check-ups. The pandemic has further encouraged the use of digital solutions like telemedicine. There has also been significant private equity investment in healthcare, with funds flowing into both multi-specialty and single-specialty hospitals and clinics.

Despite these advancements, India’s healthcare infrastructure still falls short of global standards, creating significant opportunities for development. Improvements are needed to make healthcare more efficient, accessible, and equitable across the country.

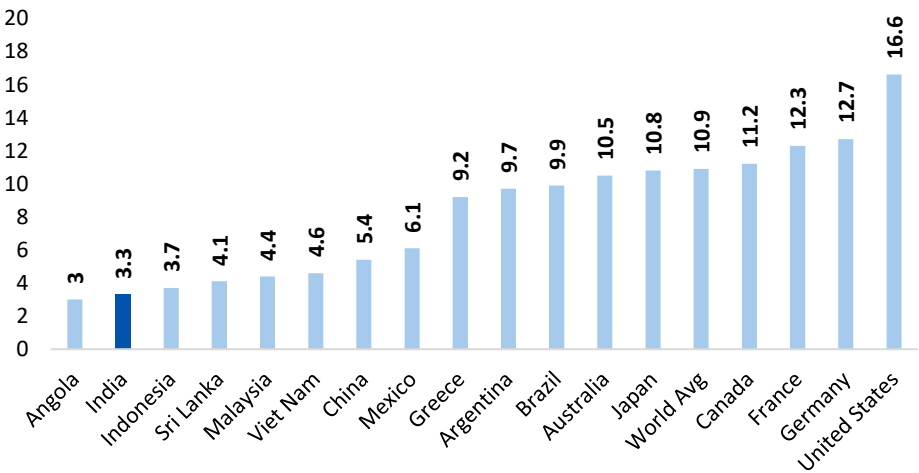
Source: Company analysis , IBEF, Niti Aayog

Healthcare Market (in billion USD)



Source: Statista

Healthcare Expenditure (% of GDP,CY22)



Source: WHO

Overview

India’s hospital industry makes up 80% of the total healthcare market and is attracting strong interest from both international and domestic investors. This market has been growing steadily, thanks to factors like rising incomes, better healthcare awareness, and a growing population. In 2023, India’s hospital market was valued at USD 98.98 billion, and it is projected to grow at 8.0% annually, reaching around USD 193.59 billion by 2032.

While large cities in India have world-class hospitals, healthcare companies are now expanding into smaller cities and towns (tier-2 and tier-3). India has a low number of hospital beds, with only 15 beds for every 10,000 people, much lower than the global average of 29 beds. This lack of basic healthcare facilities is both a challenge and an opportunity for growth. The healthcare delivery market, which includes hospital services, is expected to reach USD 193.59 billion by 2032 due to growing demand for inpatient and outpatient services.

The Indian government plans to raise public health spending to 2.5% of the GDP by 2025, which will benefit hospitals. There is a strong demand for specialized hospitals, especially in tier-2 and tier-3 cities where incomes have been rising. These areas can also serve nearby villages and smaller towns, making them strategic locations for new hospitals.

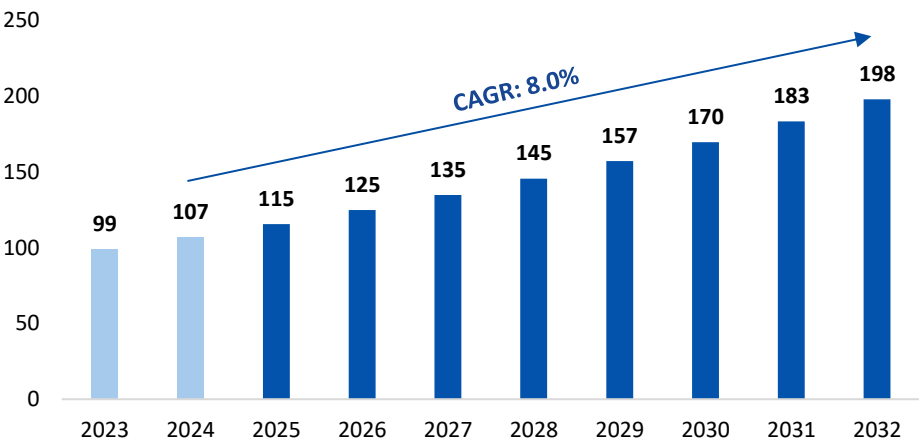
Healthcare services in India are delivered through primary, secondary, and tertiary hospitals. Primary and secondary care hospitals are fully managed by the government, while both government and private companies manage tertiary care hospitals. The private sector’s role in healthcare is growing faster than that of the government. Medical infrastructure, which includes equipment and hospital facilities, is growing at an estimated rate of 15% annually.

Source: Company analysis , IBEF, Niti Aayog ,International trade administration

Both the government and private companies are planning new specialty hospitals and upgrades to existing ones. India is expected to need 1.75 million more hospital beds by 2025, which presents an opportunity for foreign companies to invest in Indian hospitals.

The rising demand for quality healthcare, combined with limited healthcare facilities, presents both challenges and opportunities. Despite the “Make in India” initiatives, the healthcare industry still relies on imports for products like diagnostic kits, hand-held equipment, and operating room tools, with half of these products currently imported.

Hospital Market Value (USD Billion)



Source: Crisil.com

Capacity Expansion

In the last two years, the company has significant growth capex, with around ₹300 crore spent in FY2023 and ₹250 crore in FY2024. These investments reflect the company’s focus on expanding its operations and positioning itself for future growth.

For FY2025, the company has planned an ambitious capex of ₹1,600 crore. This expenditure is primarily aimed at greenfield expansions. A significant portion, ₹1,000 crore, will be used for a 1,000-bed hospital project in Bangalore, with the first phase estimated at ₹950 crore. In addition to this, there are plans for expansion in Kolkata, along with land acquisition in Rajarhat. The remaining ₹600 crore will be allocated toward biomedical maintenance (₹300 crore) and a new Cayman facility (₹250 crore), further reflecting the company's growth ambitions.

The funding for the ₹1,600 crore capex is structured with 20% coming from internal accruals and 80% from debt. As of now, the company’s gross debt, including its operations in India and the Cayman Islands, stands at ₹1,400 crore. With an additional ₹1,200 crore of planned borrowings, the total gross debt will rise to ₹2,400 crore. In a worst-case scenario, the net debt is estimated to be around ₹1,200 crore. Furthermore, the company is exploring opportunities to invest its Cayman funds outside of India.

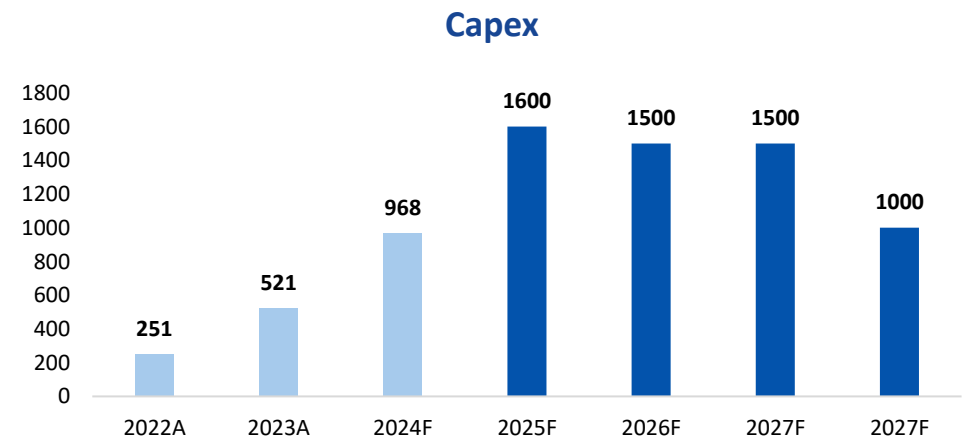
Based on the transcript from Narayana Hrudayalaya's Q1 FY25 earnings call, there was no explicit number provided for the exact capex over the next 2-3 years. However, Sandhya J., the Group CFO, mentioned that the company has indicated a certain capex for the current year, and they expect to spend a similar amount over the next 2-3 years.

Source: Company analysis

Major expansions are underway in Bangalore and Kolkata, with new hospitals and increased bed capacity. Management stated that these expansions and capital expenditures will support growth. The main turning point is expected in the next 2-3 years, once the additional capacity becomes operational. Until then, the focus will remain on improving efficiency.

The company is experiencing strong financial performance and growth in various segments, including integrated care and insurance. The planned expansions and digital transformation initiatives are expected to contribute to future topline growth, despite current challenges and temporary market conditions.

During the terminal year, capital expenditure is expected to match depreciation (100% maintenance), signaling a period of moderate but stable growth



Source: Company analysis

Government Initiative

In the 2024-25 budget, India's healthcare spending was increased to 90,659 crore from 80,518 crore in 2023-24. This shows the government's commitment to healthcare, with more funds for various health programs. The Indian government is working to improve healthcare resources and make them more accessible and affordable. According to the World Bank, India has 0.9 doctors per 1,000 people. To strengthen healthcare infrastructure, the government is planning a credit program worth Rs. 50,000 crore (about USD 6.8 billion).

The government has also announced plans to open more medical colleges using existing hospital infrastructure. This will help address the shortage of healthcare workers and increase access to medical education. The Pradhan Mantri Ayushman Bharat Health Infrastructure Mission (PM-ABHIM) aims to improve primary healthcare by establishing 14,500 Health and Wellness Centers (HWCs) across the country. These centers will offer services like preventive care, basic treatments, diagnostics, and essential medicines.

The budget also includes funding to upgrade government medical colleges, which will help increase the number of qualified healthcare professionals, especially in underserved areas. This initiative is aimed at improving access to quality healthcare and overall health outcomes in these regions.

Additionally, the decision by the Insurance Regulatory and Development Authority of India (IRDAI) to remove the age limit for buying health insurance is expected to increase health insurance coverage and make healthcare more accessible for older adults.

Source: Company analysis, Ministry of Health and Family Welfare

Technology and Digitalization

The Indian government and private sector are working together to develop advanced medical devices and digital health infrastructure. With more people using smartphones and the Internet, and with programs like the government's National Digital Health Mission, India's healthcare system is quickly moving toward digitization.

Artificial intelligence (AI) is playing an important role in healthcare. AI is helping doctors make earlier diagnoses, create personalized treatment plans, and improve patient outcomes. It provides powerful data analysis tools that support better healthcare decisions.

The digital health trend is picking up speed. Many healthcare companies are using digital tools to improve patient and doctor communication, speed up research and development, and make supply chains more efficient. These digital solutions are helping to address the shortage of healthcare professionals and are transforming healthcare in India and worldwide.

India's remote healthcare industry will also be important in the future. According to a market report, the global market for remote healthcare is expected to reach about USD 57.1 billion by 2030. AI will play a key role in this growth. Digital initiatives in India, such as the Ayushman Bharat Digital Mission, CoWIN App, Aarogya Setu, e-Sanjivani, and e-hospital, have made healthcare services available even in remote areas.

Key technology trends are changing healthcare. Big data helps improve patient care, and AI allows for better, personalized treatments. Robots handle simple tasks so doctors can focus on patients. Blockchain keeps data secure, and cloud technology makes health data easier to access. E-pharmacies are growing fast, making it easy to get medicines online.

Source: Company analysis, IBEF

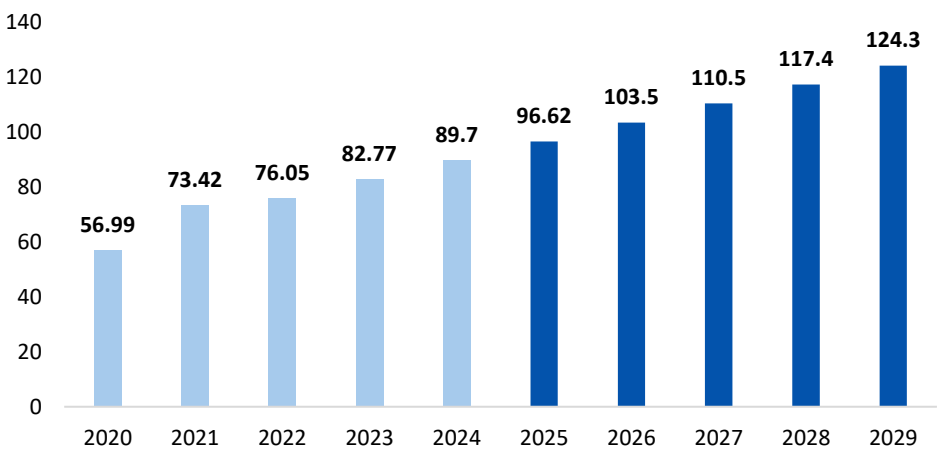
Health Insurance Penetration

The Health insurance market in India is expected to witness significant growth in the coming years. According to projections, the market size (gross written premium) is set to reach INR US\$89.70bn by 2024. This indicates a positive trend and highlights the potential for expansion within the sector.

CRISIL Research believes that although low penetration poses a significant challenge, it also offers a substantial opportunity for the expansion of the healthcare delivery sector in India. Additionally, the PMJAY scheme is anticipated to significantly enhance insurance coverage across the nation.

Moreover, as health insurance coverage in India rises, hospitalization rates are projected to increase. Mandatory health check-ups, integral to health insurance coverage, are also expected to rise, thereby driving the demand for a strong healthcare delivery infrastructure.

Health insurance market (USD Billion)



Source: Statista, Company analysis, IBEF, Crisil

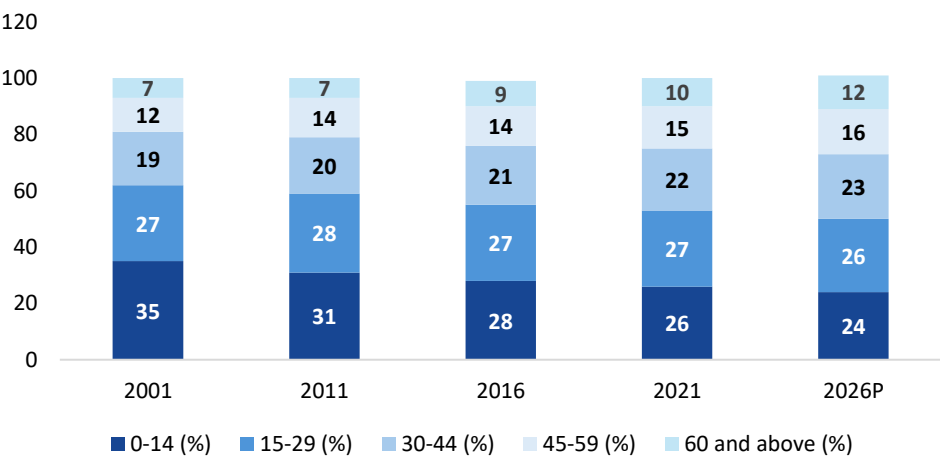
Demographics and Rising Income Levels

India's healthcare industry is set to grow, driven by the country's large population and a rising number of working-age people. By 2026, 12.5% of India's population is expected to be 67 or older. Common health issues like arthritis, hypertension, diabetes, asthma, and heart disease are likely to affect people over 60, with about 66% of the elderly experiencing at least one of these conditions.

In the next 15 years, around 15% of India's population, or about 220 million people, will be 60 or older. This increase in the senior population shows a rising need for better healthcare services tailored to their needs, supporting growth in the healthcare industry.

Additionally, as household incomes have grown over the past 15 years, more people can now access important healthcare services to stay healthy.

Indian Population by Age (%)



Source: Company Analysis, IBEF

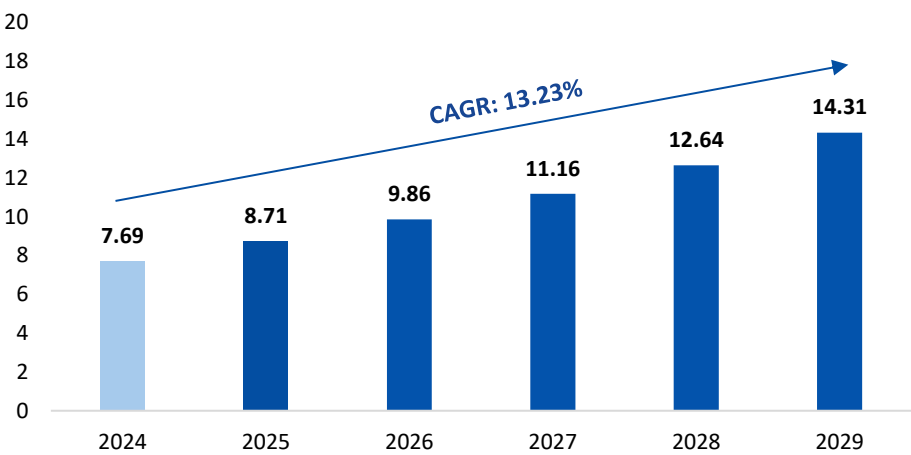
Medical Tourism

Medical tourism in India has grown a lot over the years. Each year, around 0.63 million patients from 78 countries come to India for medical, wellness, and IVF treatments, generating USD 6 billion for the industry. India's healthcare costs are affordable, especially for complex surgeries like cardiac bypasses and organ transplants. Travel and accommodation are also cheaper than in developed countries. To help international patients, the Indian government has introduced e-Medical visas and extended stays for treatment.

According to the Medical Tourism Index (MTI) 2020-2021, India ranked tenth out of 46 destinations worldwide. Most medical tourists come from South Asia, and about 94% are from Africa, West Asia, and South Asia.

The medical travel market in India was valued at USD 2.9 billion in 2020 and is expected to grow to USD 13.4 billion by 2026.

Indian Medical Tourism Market (USD Billion)



Source: Statista, Company analysis, Mordor intelligence, Ministry of Tourism

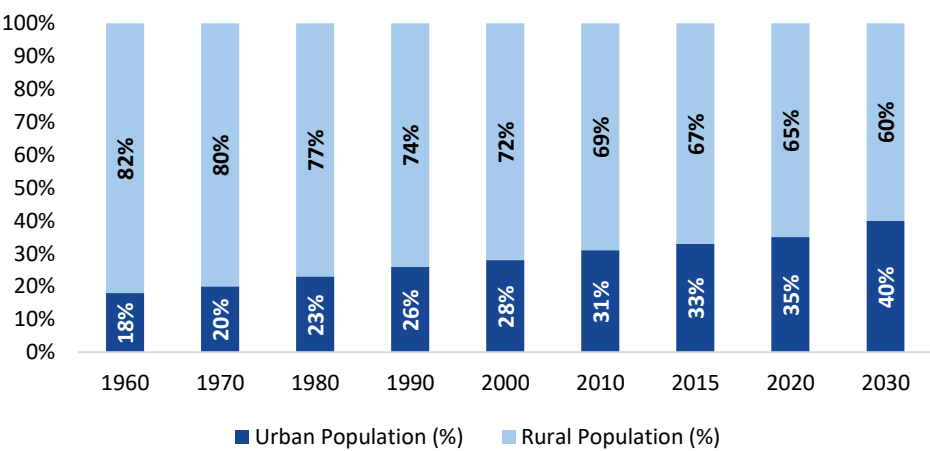
Urbanization and Growing Healthcare Awareness

Most of India's healthcare facilities are located in urban areas. With more people moving to cities, there is a growing awareness and demand for healthcare services, both for prevention and treatment, leading to higher healthcare needs.

Rising incomes and health awareness, especially after Covid, are increasing the demand for specialized and preventive care. Regular health check-ups are now more common, which has encouraged many hospital chains to expand in urban areas.

Even though urban areas have better healthcare, rural areas are also seeing more demand for basic services. This trend is likely to continue as the urban population grows.

India's urban versus rural population



Source: Company Analysis, World Urbanisation Prospects, crisil

Revenue

The company posted a revenue of INR 5,018 crores in CY2024. Over the past three years, revenue has grown at a CAGR of 24.8%, and at 11.9% over the five last year. Management has focused on improving efficiencies and optimizing the utilization of existing capacity through a combination of throughput increases, payer mix optimization, and digital initiatives. These efforts aim to sustain double-digit growth, supported by plans for future expansion.

The company has optimized its bed mix by upgrading to single and twin-sharing rooms, which has generated higher-quality revenue within the same patient volume. They expect steady double-digit revenue growth. However, the Current Growth will be normal growth and the inflection point will come when additional capacity becomes operational in the next 2-3 years. Until then, the focus will be on improving efficiencies.

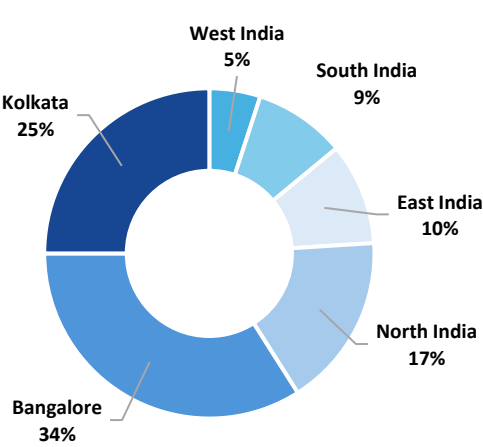
As stated by management, “The growth will be organic in the short term. The next significant inflection point will occur when company additional capacities come online. Until then, growth will align with the market and with company's internal efforts focused on throughput maximization and improving revenue quality.”

Management has guided for green field expansion of 550 beds which will be fully operational post FY’27, revenue and profits from it will commence in FY’28 And new facility at Cayman island (which is at heart of city) gains traction. Company has capex plan for next 2-3 years expected to be around 4000 Crores.

The analyst censess growth for FY25 INR 5721 Crores which is 14% over the FY24 revenue of INR 5018 crores. The company is signaling that the major growth will occur when the new capacities, specifically the new facilities in Kolkata and Bangalore, become operational.

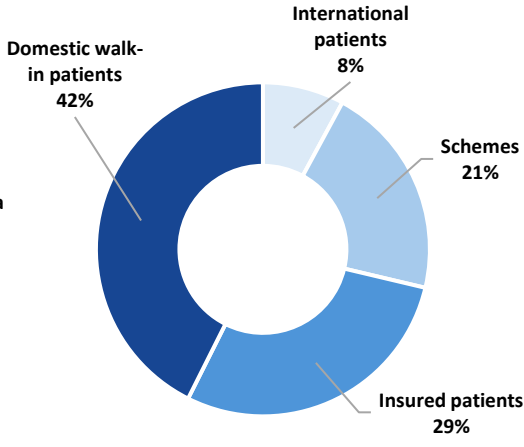
Source: Company Analysis

Location Wise Break-up

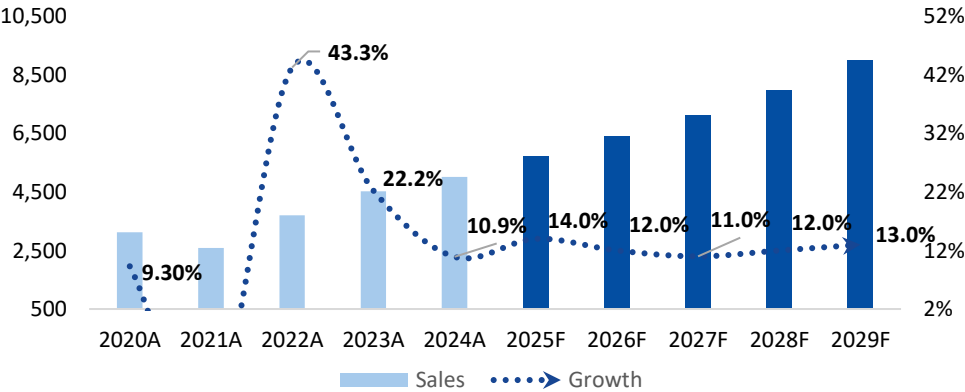


Source: Tijori.com

Customer Segment



Revenue Growth



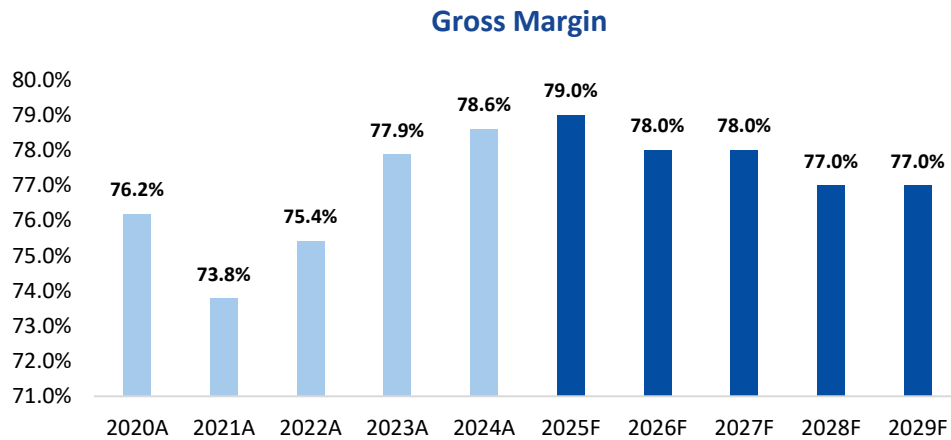
Source: Company Analysis, Tijori.com

Gross Margin

The company's gross margins have remained relatively stable, ranging between 75.8% and 75.6%, while the cost of medicines and consumables has consistently accounted for approximately 24.0% to 25.0% of revenue. Over the past two years, gross margins increased by 2-3%, primarily due to a reduction in material . However, in the coming years, margins are expected to normalize within their historical range.

Historically, the company has posted gross margins between 74.3% and 78.6%, with a median of 75.8% over the past decade. Based on our research, the gross margins for hospitals and healthcare facilities in India averaged 46.06%.

In the global emerging market healthcare sector, margins were 39.54%, while the global market average was 34.07%.



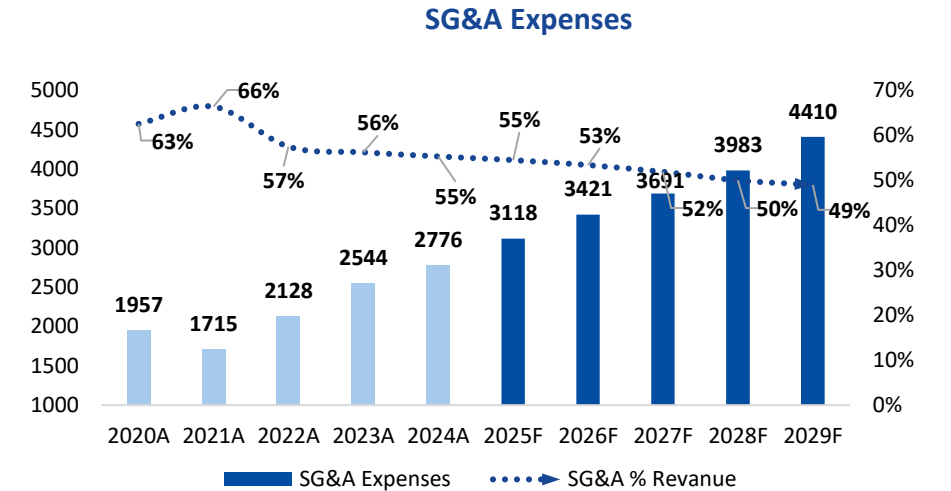
Source: Company Analysis

SG&A Expenses

During FY24, Company has reported EBITDA margin of 23.3% Which is 200bps higher than FY23(21.6%). The reason for rise in EBITDA margin Because of reduce the misc. cost. during FY23 and FY24 EBITDA margin increase largely because of reduce the martial cost .

Basis our research, we have identified that historically,SG&A expenses is 63.8% of revenue. Where few expenses viz Employ cost, Selling and Administration cost, Other Misc. Expenses Constitute 59.7% of Total SG&A expenses. these five key expenses are directly linked with sales.

We assumed ,total SG&A expenses to remain at 60.0% level as percentage of revenue which is almost align with street estimate and management guidance.



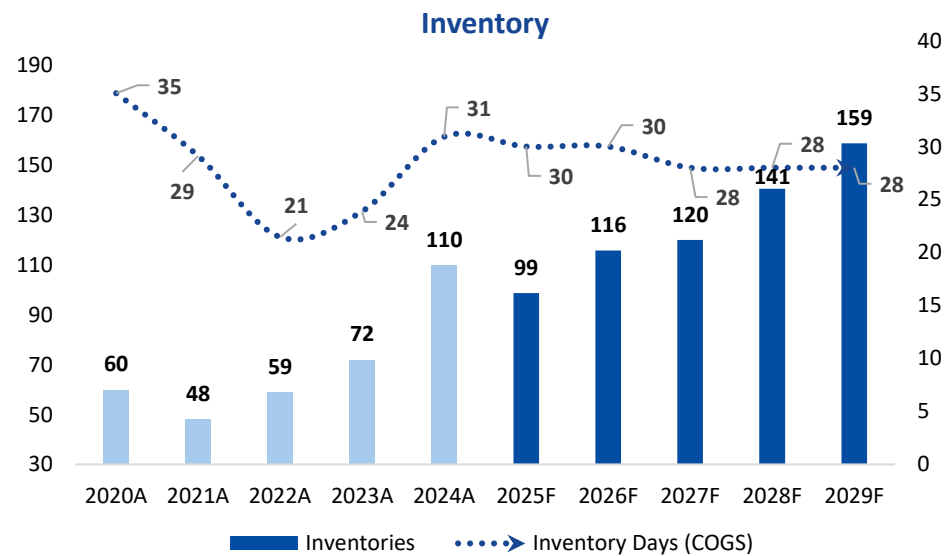
Source: Company Analysis

Inventory

Historically, the company has maintained inventory levels at approximately 2.7% of revenue, with a median inventory turnover ratio of 8.55x. Over the past 3-4 years, the inventory turnover ratio has increased from around 6x to 9x, resulting in inventory days of 43. In FY24, inventory levels reached their highest point in the past 10 years, primarily due to stockpiles of medical consumables, drugs, and surgical instruments.

In comparison, peer companies in the market maintain inventory levels of around 30 days, which is lower than the company’s 10-year historical inventory days.

Furthermore, street estimates suggest that inventory levels will reduce to 30-35 days from the current levels.

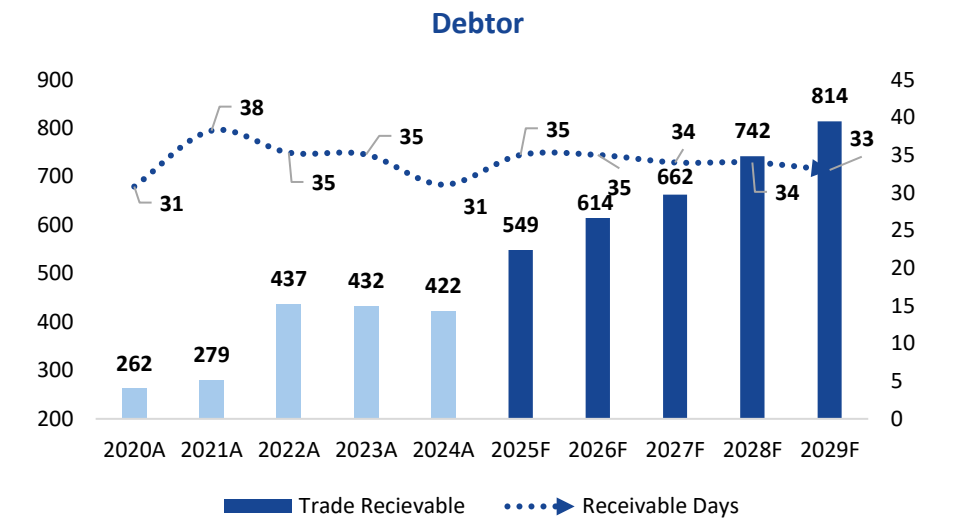


Source: Company Analysis

Debtor

The company has a steady receivables process , which ensures that receivable levels remain within normal ranges. As of March 31, 2024, 50.8% of receivables were due for less than six months. The company is exposed to credit risk from its operating activities (trade receivables) and investing activities. However, this risk is controlled by continuously analyzing the credit limits and creditworthiness of customers.

Historically, the company has maintained receivables at approximately 9.6% of revenue. The median receivable turnover ratio is 10.47x, translating to 35 receivable days, which is in line with peers in the market who maintain receivable days between 32-36 days. This also aligns with the company's 10-year historical average of 35 days. Street estimates suggest that receivable levels are expected to remain flat, with consensus estimates projecting receivable days to remain steady at 35 days. Given this historical consistency and industry alignment, we assume that the company's receivable levels will remain flat at 35 days going forward.



Source: Company Analysis

Payables

Comany has maintained consistent payable of ~180-200 days. in year 2024, the payables were reported to normal level of 205 days. The trade payables is Constantly ~49% of COMC and ~11.9% Revenue.

The peers companies in the market maintains payables of 150 Days which is Lower than 10 year historical Payable days of the company. Further , street estimates suggest the payable levels to ~200-250 suggest.

In absence of management guidance and basis our research, we assume the payable days to remain flat at 180 days

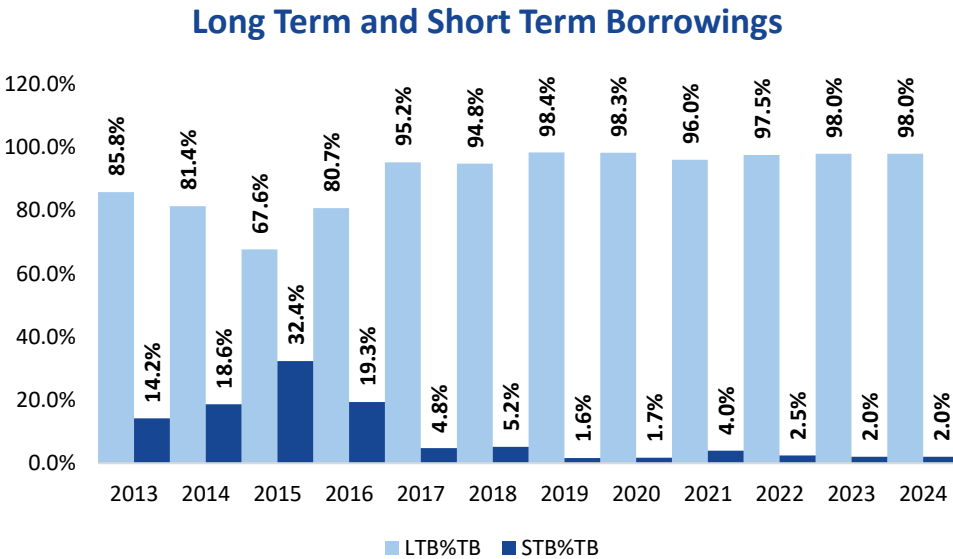
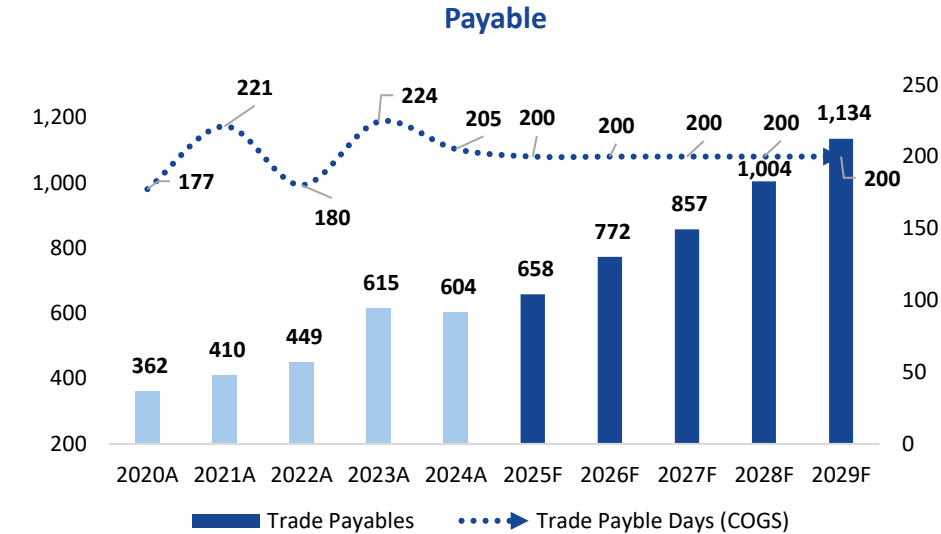
Borrowings

Over the past two years, the company’s long-term debt has grown to around 98% of total debt in 2024, up from a 10-year median of 95%. Financial ratios remain stable, with a Debt to Equity ratio of 0.4x, Debt to Capital at 0.3x, Debt to Assets at 0.2x, and Debt to EBITDA at 1.0x. While the company targets a Debt to EBITDA ratio of 2, it may exceed this due to investments in greenfield projects, such as land acquisition and new hospitals.

The management plans annual capex of ₹1,500–1,600 crore over the next three years, with 80% funded by debt. Although the 20%-80% internal accruals-to-debt mix is a conservative estimate, cash accruals will be prioritized over debt. Debt levels are expected to remain elevated for the next 2-3 years until new investments generate cash flows.

During this period, long-term debt may grow by ~22-24% of revenue next 2-3 years, while short-term debt is expected to increase by ~0.7% of revenue.

Source: Company Analysis



Source: Company Analysis

Objective

The objective is to reach at the intrinsic value of Shree cement stock. The analysis aim to assess the company's worth basis its ability to generate cash flows, future growth and risk associated with the business. We have carefully considered management's guidance while performing this analysis.

Valuation Methodology

There are primarily three approach in valuation (Viz., Cost Approach, Market Approach, and Income Approach)

For any valuation, all the approaches may not be relevant and therefore will not give a fair estimate of value. Hence, the approach most suitable for specific business/company must be applied in the valuation exercise based on common practices.

The three approaches generally adopted in valuation are as under:

- Market Approach: Comparable Companies Method and Comparable Transaction Method.
- Income Approach : Discounted Cash Flow Method
- Asset Approach : Net Asset Value method.

Market Approach

Under this approach, we measure the value of business of the company by applying the market multiple of listed companies which trade actively and possess attributes similar and comparable with the target company. This methodology is based on the principle that such market pricing taking place between informed buyers and informed

Seller while taking into account all the relevant factors. Not all multiple are applicable to every company. Therefore, the relevant multiple should be chosen carefully and adjusted for differences between the comparable. For the purpose of valuation of Shree cement, we have chosen EV/EBITDA multiple.

Asset Approach

Under this approach, the value of the business is derived basis difference between the value of asset and liabilities. This approach focusses on determining the value of net assets from the perspective of equity valuation. The value of net asset can be determined as follows; $\text{Net assets} = \text{Total Asseer} - \text{Total Liabilities}$. The value of net assets is also known as Total Equity.

Income Approach

Under this approach, The value of business is derived by considering free cash flows, future growth and weighted average cost of capital which signifies risk.

There are four major steps in income Approach.

- Forecasting future cash flow for explicit growth period.
- Discounting these cash flow to present value at the rate of return which despite the risk in realizing the future cash flow and expected growth.
- Calculation of terminal value of free cash flows post explicit growth period.
- Addition of present value of terminal cash flow and free cash flow during explicit forecast period.

Annexure-1

WACC Calculation

WACC Inputs	
Country	India
Local Currency 10Y Government Bond Yield	6.82%
India's Moody's Sovereign Rating	Baa3
Credit default spread as per Dr.Damodaran	2.4%
Implied US Equity Risk Premium	3.96%
Asset Beta	0.45
Asset Beta Variation (+/-)	0.05
Debt to Equity Ratio	42.3%
Debt to Capital (Gearing)	29.7%
Marginal Tax Rate	25%
Company Credit Rating Available?	Yes
Company is Large Firm or small Firm	L
Provide Credit Rating	Aa2/AA
Interest Coverage Ratio	8.27
ICR Based Corporate Default Spread	0.70%
Small Size Premium	5.30%

Particulars	Low	Mid	High
Risk Free Rate	4.43%	4.43%	4.43%
Equity Risk Premium	3.96%	3.96%	3.96%
Asset Beta	0.40	0.45	0.50
D/E Ratio	42.3%	42.3%	42.3%
Equity Beta	0.53	0.59	0.66
CAPM Cost of Equity	6.51%	6.77%	7.03%

Small Size Premium	5.30%	5.30%	5.30%
Asset Specific Premium	0.00%	0.00%	0.00%
Modified Cost of Equity	11.81%	12.07%	12.33%

Pre Tax Cost of Debt	7.52%	7.52%	7.52%
Marginal tax Rate	25%	25%	25%
Post Tax Cost of Debt	5.64%	5.64%	5.64%

Capital Gearing (D/D+E)	29.73%	29.73%	29.73%
Weighted Average Cost Of Capital	9.98%	10.16%	10.34%
WACC (Rounded)	10.00%	11.00%	11.00%

Risk Free Rate

We have considered 4.59% as the risk free rate in estimation of cost of equity of 7.26%.The risk free rate of India is calculated by considering local currency 10 Year government bond yield adjusted for default spread of India.

Equity Risk Premium

An ERP of 6.51% has been adopted based on US implied equity risk premium, relative equity volatile of ICE BOFA public sector issue Emerging Markets index, S&P Emerging BMI Index and Country risk Premium of India.

Asset Beta

An ERP of 3.96% has been adopted based on US implied equity risk premium, relative equity volatile of ICE BOFA public sector issue Emerging MKts index, S&P Emerging BMI Index and Country risk Premium of India.

Cost of Debt

For the purpose of assessing the cost of debt, we have used country's risk free rate, country's default spread and corporate default spread based on company's debt rating and interest coverage ratio.

Small Size Premium

Empirically small sized assets have exceeded the returns as compared to large sized assets. We have used the study on size premium conducted by incwert which is based on difference between actual returns and CAPM expected returns arranged in decile basis market capitalization of Indian companies.

Notes :

- 1.Tax rate considered as marginal tax rate of India.
2. Asset Beta is considered as median of 2 years beta of comps adjusted for Blume affect.
3. Equity Beta =Assets Beta * (1+(1-Tax Rate%)*Debt/Equity)

Beta Analysis 2 Years Weekly

Comparable Company	No. of Weekly Observation	Market Cap	Asset Beta	2 Year Avg D/E	Equity Beta	2 Year avg D/D+E
Apollo Hospitals	104	99,856	0.51	0.7x	0.79	0.7x
Max Healthcare	104	91,186	0.89	0.1x	0.97	0.1x
Fortis Healthcare	104	43,999	0.71	0.1x	0.79	0.1x
KIMS	104	20,917	0.16	0.6x	0.23	0.6x
Global Health	104	28,470	0.24	0.4x	0.31	0.4x
Rainbow Medicare	104	14,428	0.08	0.6x	0.11	0.6x
Aster DM Healthcare	104	22,416	0.32	1.0x	0.55	1.0x
Median - Broad set			0.32	0.6x	0.55	0.6x
Average - Broad set			0.42	0.5x	0.54	0.5x
Median - Considered set			0.61	0.4x	0.79	0.4x
Average - Considered set			0.61	0.5x	0.77	0.5x

Considered?	Narrow Set
Y	TRUE
Y	TRUE
Y	TRUE
N	FALSE
N	FALSE
N	FALSE
Y	TRUE

Blume Beta Analysis 2 Years Weekly

Comparable Company	No. of Weekly Observation	Market Cap	Asset Beta	2 Year Avg D/E	Equity Beta	2 Year avg D/D+E
Apollo Hospitals	104	99,856	0.53	0.7x	0.83	0.7x
Max Healthcare	104	91,186	0.86	0.1x	0.95	0.1x
Fortis Healthcare	104	43,999	0.75	0.1x	0.83	0.1x
KIMS	104	20,917	0.33	0.6x	0.48	0.6x
Global Health	104	28,470	0.41	0.4x	0.53	0.4x
Rainbow Medicare	104	14,428	0.28	0.6x	0.40	0.6x
Aster DM Healthcare	104	22,416	0.39	1.0x	0.68	1.0x
Median - Broad set			0.41	0.6x	0.68	0.6x
Average - Broad set			0.51	0.5x	0.67	0.5x
Median - Considered set			0.64	0.44	0.83	0.44
Average - Considered set			0.64	0.49	0.82	0.49

Considered?	Narrow Set
Y	TRUE
Y	TRUE
Y	TRUE
N	FALSE
N	FALSE
N	FALSE
Y	TRUE

Beta Analysis 5 Years Weekly

Comparable Company	No. of Weekly Observation	Market Cap	Asset Beta	5 Year Avg D/E	Equity Beta	5 Year avg D/D+E
Apollo Hospitals	60	99,856	0.36	0.8x	0.59	0.5x
Max Healthcare	60	91,186	0.20	0.4x	0.26	0.2x
Fortis Healthcare	60	43,999	0.54	0.2x	0.62	0.2x
KIMS	60	20,917	-0.09	0.5x	-0.12	0.3x
Global Health	60	28,470	0.00	0.6x	0.00	0.4x
Rainbow Medicare	60	14,428	0.20	0.9x	0.34	0.5x
Aster DM Healthcare	60	22,416	0.33	1.3x	0.66	0.5x
Median - Broad set			0.20	0.6x	0.34	0.4x
Average - Broad set			0.22	0.7x	0.34	0.4x
Median - Considered set			0.36	0.83	0.62	0.45
Average - Considered set			0.36	0.83	0.62	0.45

Considered?	Narrow Set
Y	TRUE
N	FALSE
Y	TRUE
N	FALSE
N	FALSE
N	FALSE
Y	TRUE

Blume Beta Analysis 5 Years Weekly

Comparable Company	No. of Weekly Observation	Market Cap	Asset Beta	2 Year Avg D/E	Equity Beta	2 Year avg D/D+E
Apollo Hospitals	60	99,856	0.45	0.7x	0.70	0.7x
Max Healthcare	60	91,186	0.45	0.1x	0.49	0.1x
Fortis Healthcare	60	43,999	0.65	0.1x	0.72	0.1x
KIMS	60	20,917	0.18	0.6x	0.25	0.6x
Global Health	60	28,470	0.00	0.4x	0.33	0.4x
Rainbow Medicare	60	14,428	0.38	0.6x	0.55	0.6x
Aster DM Healthcare	60	22,416	0.44	1.0x	0.75	1.0x
Median - Broad set			0.44	0.6x	0.55	0.6x
Average - Broad set			0.36	0.5x	0.54	0.5x
Median - Considered set			0.45	0.44	0.71	0.44
Average - Considered set			0.45	0.44	0.71	0.44

Considered?	Narrow Set
Y	TRUE
Y	TRUE
Y	TRUE
N	FALSE
N	FALSE
N	FALSE
Y	TRUE

We have taking 0.45 as asset beta of hospital.

3 Statement Model : Income Statement

In INR crores unless Stated otherwise

	Actual					Explicit Forecast Period				
	FY2020	FY2021	FY2022	FY2023	FY2024	FY2025	FY2026	FY2027	FY2028	FY2029
Income Statement										
Revenue	3,128	2,583	3,701	4,524	5,018	5,721	6,407	7,112	7,965	9,001
Less: Cost of Medicines and Consumables	745	677	910	1,001	1,074	1,201	1,410	1,565	1,832	2,070
Gross Profit	2,383	1,906	2,791	3,523	3,944	4,519	4,997	5,547	6,133	6,930
Gross Margin	76.2%	73.8%	75.4%	77.9%	78.6%	79.0%	78.0%	78.0%	77.0%	77.0%
Less: SG&A Expenses (Excluding Depreciation & Interest)										
Employee Cost	674	627	768	879	1,004	1,144	1,281	1,422	1,513	1,710
Operation Costs	897	778	983	1,182	1,282	1,430	1,538	1,636	1,832	1,980
Power and Fuel Cost	91	84	97	106	118	137	147	164	159	180
Selling, Administration and Other Cost	268	211	241	338	374	400	448	462	478	540
Misc. Cost	27	15	39	39	(2)	6	6	7	-	-
Earning Before Interest Tax & Depreciation & Amortization	426	191	663	979	1,168	1,402	1,576	1,856	2,151	2,520
EBITDA MARGIN %	13.6%	7.4%	17.9%	21.6%	23.3%	24.5%	24.6%	26.1%	27.0%	28.0%
Less: Depreciation & Amortization	186	184	183	210	242	342	436	530	592	624
Earning Before Interest & Tax	240	7	480	769	926	1,060	1,140	1,327	1,559	1,896
EBIT MARGIN %	7.7%	0.3%	13.0%	17.0%	18.5%	18.5%	17.8%	18.7%	19.6%	21.1%
Less : Interest	94	82	76	84	112	107	115	114	89	44
Add : Other Income	28	27	35	65	75	66	74	82	92	-
Exceptional Income/Expenses	(13)	(8)	(9)	2	-	-	-	-	-	-
Profit Before Tax	161	(56)	430	752	889	1,019	1,100	1,295	1,562	1,852
Less: Taxes	42	(42)	88	145	99	102	110	324	391	556
Net Income	119	(14)	342	607	790	917	990	971	1,172	1,296

3 Statement Model : Balance Sheet

In INR crores unless Stated otherwise

	Actual					Explicit Forecast Period				
	FY2020	FY2021	FY2022	FY2023	FY2024	FY2025	FY2026	FY2027	FY2028	FY2029
Balance Sheet										
Assets										
Current Assets										
Cash & Cash Equivalents	115	132	172	379	417	160	145	126	259	912
Inventory	60	48	59	72	110	99	116	120	141	159
Debtors(Trade Receivable)	262	279	437	432	422	549	614	662	742	814
Loan	45	46	70	108	102	102	102	102	102	102
Investment	69	110	131	251	840	840	840	840	840	840
Other Current Assets	32	48	45	50	55	74	83	78	80	90
Total Current Assets	583	663	914	1,292	1,946	1,824	1,901	1,929	2,163	2,917
Non-Current Assets										
Net Property, Plant & Equipment	1,725	1,644	1,707	1,923	2,198	3,488	4,584	5,587	6,027	5,943
Right of Use Assets	237	200	169	131	202	183	163	144	125	106
Intangible Assets	72	58	46	166	133	120	108	95	82	69
Good Will	58	58	58	117	118	118	118	118	118	118
Capital Work in progress	12	18	63	259	514	514	514	514	514	514
Non Current Financial Assets	199	122	173	267	508	508	508	508	508	508
Total Non-Current Assets	2,303	2,100	2,216	2,863	3,673	4,931	5,995	6,966	7,374	7,258
Total Assets	2,886	2,763	3,130	4,155	5,619	6,755	7,896	8,895	9,537	10,175
Equity and Liabilities										
Equity										
Share Capital	204	204	204	204	204	204	204	204	204	204
Other Equity	928	912	1,281	1,926	2,679	3,596	4,586	5,557	6,611	7,649
Total Shareholder's Equity	1,132	1,116	1,485	2,130	2,883	3,800	4,790	5,761	6,815	7,853
Minority Interest	-	-	-	-	-	-	-	-	-	-
Total Equity	1,132	1,116	1,485	2,130	2,883	3,800	4,790	5,761	6,815	7,853
Non-Current Liabilities:										
Long-Term Borrowings	619	504	432	626	1,195	1,373	1,410	1,351	797	270
Other Non-Current Liabilities:	532	440	428	410	429	429	429	429	429	429
Deferred Tax Liabilities (Net)	52	11	50	81	42	42	42	42	42	42
Other Long Term Liabilities	460	404	348	294	346	346	346	346	346	346
Long Term Provisions	20	25	30	35	41	41	41	41	41	41
Total Non-Current Liabilities	1,151	944	860	1,036	1,624	1,802	1,839	1,780	1,226	699
Current Liabilities:										
Trade Payables	362	410	449	615	604	658	772	857	1,004	1,134
Short-Term Borrowings	11	21	11	13	25	11	12	13	9	6
Other Current Liabilities	230	272	325	361	483	483	483	483	483	483
Short Term Provisions	33	30	45	53	73	73	73	73	73	73
Non controlling Interest	1	1	1	1	1	1	1	1	1	1
Other Current Liabilities	196	241	279	307	409	409	409	409	409	409
Total Current Liability	603	703	785	989	1,112	1,153	1,268	1,353	1,496	1,623
Total Equity and Liabilities	2,886	2,763	3,130	4,155	5,619	6,755	7,896	8,895	9,537	10,175

3 Statement Model : Cash Flow Statement

In INR crores unless Stated otherwise

	Actual					Explicit Forecast Period				
	FY2020	FY2021	FY2022	FY2023	FY2024	FY2025	FY2026	FY2027	FY2028	FY2029
Cash Flow Statement										
Profit Before Tax	161	(56)	430	752	889	1,019	1,100	1,295	1,562	1,852
Add : Interest Expenses	85	76	66	69	97	107	115	114	89	44
Add : Depreciation	186	184	183	210	242	342	436	530	592	624
Add : Other items	24	(10)	25	35	(55)	-	-	-	-	-
Less : Change in Working capital	53	56	(152)	105	(15)	(80)	22	38	45	30
Less : Tax Paid	(66)	54	(67)	(86)	(91)	(102)	(110)	(324)	(391)	(556)
Cash Flow from Oration	443	304	485	1,085	1,067	1,286	1,563	1,653	1,898	1,995
Net Purchase of Fixed Assets	(115)	(70)	(251)	(521)	(968)	(1,600)	(1,500)	(1,500)	(1,000)	(508)
Net Purchase of Investments	(67)	(42)	(25)	(201)	(585)	-	-	-	-	-
Addition of CWIP	-	-	-	-	-	-	-	-	-	-
Interest Received	1	8	1	8	18	-	-	-	-	-
Acquisition of companies	-	-	-	(43)	-	-	-	-	-	-
Other investing activities items	(8)	(9)	6	(419)	77	-	-	-	-	-
Cash Flow from Investing Activites	(189)	(113)	(269)	(1,176)	(1,458)	(1,600)	(1,500)	(1,500)	(1,000)	(508)
Net Proceeds from Borrowings	(108)	(102)	(68)	209	368	164	37	(58)	(559)	(530)
Dividend Paid	(49)	-	-	(20)	(51)	-	-	-	(117)	(259)
Interest Paid	(51)	(43)	(36)	(43)	(73)	(107)	(115)	(114)	(89)	(44)
Proceeds from debentures	-	-	-	-	300	-	-	-	-	-
Financial liabilities	(39)	(48)	(55)	(57)	(56)	-	-	-	-	-
Cash Flow from Financing Activites	(247)	(193)	(159)	89	488	58	(78)	(172)	(765)	(833)
Net Change In Cash	7	(2)	57	(2)	97	(257)	(15)	(19)	133	653
Effect of Foreign exchange fluctuations	-	-	-	(2)	-	-	-	-	-	-
Opening Cash Balance	84	92	90	149	145	417	160	145	126	259
Closing Cash Balance	91	90	147	145	242	160	145	126	259	912

DCF valuation

in INR Crores unless stated otherwise

	Actual					Explicit Forecast Period				
	FY2020	FY2021	FY2022	FY2023	FY2024	FY2025	FY2026	FY2027	FY2028	FY2029
Free Cash Flow to Firm										
EBITDA	426	191	663	979	1,168	1,402	1,576	1,856	2,151	2,520
Less:Depreciation & Amortization	(186)	(184)	(183)	(210)	(242)	(342)	(436)	(530)	(592)	(624)
EBIT	240	7	480	769	926	1,060	1,140	1,327	1,559	1,896
Less:Tax	(42)	42	(88)	(145)	(99)	(102)	(110)	(324)	(391)	(556)
EBIT(1-T)	198	49	392	624	827	958	1,030	1,003	1,168	1,341
Add:Depreciation & Amortization	186	184	183	210	242	342	436	530	592	624
Cash NOPAT	384	233	575	834	1,069	1,300	1,466	1,532	1,760	1,965
Less:Reinvestment										
Change in Working Capital	53	56	(152)	105	(15)	(80)	22	38	45	30
Capex during the year	(115)	(70)	(251)	(521)	(968)	(1,600)	(1,500)	(1,500)	(1,000)	(508)
Free Cash Flow To Firm	322	219	172	418	86	(381)	(12)	70	805	1,487
Discount Period(Years)	Mid Year Convention? ☐ Yes					0.5	1.5	2.5	3.5	4.5
Discount Rate (WACC)	10%					0.953	0.865	0.785	0.713	0.647
Present ValueFactor										
Present value of Free Cash Flow to Firm						(363)	(10)	55	574	962

DCF Calculation - Perpetuity Growth		DCF Calculation - Exit Multiple		Market Cap INR 25,508 Crores	WACC	Terminal Growth Rate%	3%	4%	5%	6%	7%
Perpetuity Growth Rate	4.43%	Exit Multiple	35.4 x	Intrinsic Value INR 20,373 Crores	WACC	Terminal Growth Rate%	33x	34x	35x	36x	37x
Terminal Year FCFF x (1+PGR)	1,552	Terminal Year EBITDA	2,520								
Terminal-Value	27,071	Terminal-Value	89,214	WACC 10%	WACC	Terminal Growth Rate%	33x	34x	35x	36x	37x
Present Value of Terminal Value	17,514	Present Value of Terminal Value	57,719								
Present Value of FCFFs	1,218	Present Value of FCFFs	1,218	Terminal Growth 4.43%	WACC	Terminal Growth Rate%	33x	34x	35x	36x	37x
Enterprise Value	18,732	Enterprise Value	58,937								
Less: Debt	1,220	Less:Debt	1,220		WACC	Terminal Growth Rate%	33x	34x	35x	36x	37x
Add: Cash	417	Add: Cash	417								
Less: Minority Interest	-	Less:Minority Interest	-		WACC	Terminal Growth Rate%	33x	34x	35x	36x	37x
Add:Non Operating Asset	4	Add:Non Operating Asset	4								
Equity Value	20,373	Equity Value	60,578		WACC	Terminal Growth Rate%	33x	34x	35x	36x	37x
Less: Value of Options	-	Less: Value of Options	-								
Equity Value for Shareholder	20,373	Equity Value for Shareholder	60,578								

Ratio Analysis

Explicit Forecast Period

in INR Crores unless stated otherwise

	FY2020	FY2021	FY2022	FY2023	FY2024	FY2025	FY2026	FY2027	FY2028	FY2029
Liquidity Ratio										
Current Assets	583	663	914	1,292	1,946	1,824	1,901	1,929	2,163	2,917
Current Liabilities	603	703	785	989	1,112	1,153	1,268	1,353	1,496	1,623
Current Ratio	1.0x	0.9x	1.2x	1.3x	1.8x	1.6x	1.5x	1.4x	1.4x	1.8x
Current Assets	583	663	914	1,292	1,946	1,824	1,901	1,929	2,163	2,917
Inventory	60	48	59	72	110	99	116	120	141	159
Current Liabilities	603	703	785	989	1,112	1,153	1,268	1,353	1,496	1,623
Quick Ratio(Acid Test)	0.9x	0.9x	1.1x	1.2x	1.7x	1.5x	1.4x	1.3x	1.4x	1.7x
Inventory Days	35 Days	29 Days	21 Days	24 Days	31 Days	30 Days	30 Days	28 Days	28 Days	28 Days
Debtor Days	31 Days	38 Days	35 Days	35 Days	31 Days	35 Days	35 Days	34 Days	34 Days	33 Days
Creditor Days	177 Days	221 Days	180 Days	224 Days	205 Days	200 Days	200 Days	200 Days	200 Days	200 Days
Cash Conversion Days	(112)	(154)	(123)	(165)	(143)	(135)	(135)	(138)	(138)	(139)
Leverage Ratio/Solvency Ratio										
Total Debt	630	525	443	639	1,220	1,384	1,422	1,364	806	276
Total Equity	1,132	1,116	1,485	2,130	2,883	3,800	4,790	5,761	6,815	7,853
Debt to Equity Ratio	0.6x	0.5x	0.3x	0.3x	0.4x	0.4x	0.3x	0.2x	0.1x	0.0x
Total Debt	630	525	443	639	1,220	1,384	1,422	1,364	806	276
Total Assets	2,886	2,763	3,130	4,155	5,619	6,755	7,896	8,895	9,537	10,175
Debt to Assets Ratio	0.2x	0.2x	0.1x	0.2x	0.2x	0.2x	0.2x	0.2x	0.1x	0.0x
Total Equity	1,132	1,116	1,485	2,130	2,883	3,800	4,790	5,761	6,815	7,853
Total Assets	2,886	2,763	3,130	4,155	5,619	6,755	7,896	8,895	9,537	10,175
Equity Ratio/Multiplayer	0.4x	0.4x	0.5x	0.5x	0.5x	0.6x	0.6x	0.6x	0.7x	0.8x
Coverage Ratio										
Total Debt	630	525	443	639	1,220	1,384	1,422	1,364	806	276
EBITDA	426	38	35	35	31	35	35	34	34	33
Debt to EBITDA Coverage Ratio	1.5x	13.7x	12.5x	18.2x	39.3x	39.6x	40.6x	40.1x	23.7x	8.4x
EBIT	240	7	480	769	926	1,060	1,140	1,327	1,559	1,896
Interest	94	82	76	84	112	107	115	114	89	44
Debt Repayment	105	82	(196)	(581)	(164)	(37)	58	559	530	276
Debt Service Coverage Ratio	1.2x	0.0x	-4.0x	-1.5x	-17.7x	15.3x	6.6x	2.0x	2.5x	5.9x
EBIT	240	7	480	769	926	1,060	1,140	1,327	1,559	1,896
Interest Expenses	94	82	76	84	112	107	115	114	89	44
Interest Coverage Ratio	2.6x	0.1x	6.3x	9.2x	8.3x	9.9x	9.9x	11.6x	17.5x	42.8x

Ratio Analysis

in INR Crores unless stated otherwise

						Explicit Forecast Period				
	FY2020	FY2021	FY2022	FY2023	FY2024	FY2025	FY2026	FY2027	FY2028	FY2029
Profitability/Margins ratio										
Gross Profit	2,383	1,906	2,791	3,523	3,944	4,519	4,997	5,547	6,133	6,930
Total sales	3,128	2,583	3,701	4,524	5,018	5,721	6,407	7,112	7,965	9,001
Gross Margin	76.2%	73.8%	75.4%	77.9%	78.6%	79.0%	78.0%	78.0%	77.0%	77.0%
EBITDA	426	191	663	979	1,168	1,402	1,576	1,856	2,151	2,520
Total sales	3,128	2,583	3,701	4,524	5,018	5,721	6,407	7,112	7,965	9,001
EBITDA Margin	13.6%	7.4%	17.9%	21.6%	23.3%	24.5%	24.6%	26.1%	27.0%	28.0%
EBIT	240	7	480	769	926	1,060	1,140	1,327	1,559	1,896
Total sales	3,128	2,583	3,701	4,524	5,018	5,721	6,407	7,112	7,965	9,001
EBIT Margin	7.7%	0.3%	13.0%	17.0%	18.5%	18.5%	17.8%	18.7%	19.6%	21.1%
Net Profit	119	(14)	342	607	790	917	990	971	1,172	1,296
Total sales	3,128	2,583	3,701	4,524	5,018	5,721	6,407	7,112	7,965	9,001
Net Profit Margin	3.8%	-0.5%	9.2%	13.4%	15.7%	16.0%	15.4%	13.7%	14.7%	14.4%
Net Income	119	(14)	342	607	790	917	990	971	1,172	1,296
Total Assets	2,886	2,763	3,130	4,155	5,619	6,755	7,896	8,895	9,537	10,175
Return on Assets	4.1%	-0.5%	10.9%	14.6%	14.1%	13.6%	12.5%	10.9%	12.3%	12.7%
Net Income	119	(14)	342	607	790	917	990	971	1,172	1,296
Total Equity	1,132	1,116	1,485	2,130	2,883	3,800	4,790	5,761	6,815	7,853
Return on Equity	10.5%	-1.3%	23.0%	28.5%	27.4%	24.1%	20.7%	16.9%	17.2%	16.5%
Efficiency Ratio										
Revenue	3,128	2,583	3,701	4,524	5,018	5,721	6,407	7,112	7,965	9,001
Total Asset	2,886	2,763	3,130	4,155	5,619	6,755	7,896	8,895	9,537	10,175
Asset Turnover Ratio	1.1x	0.9x	1.2x	1.1x	0.9x	0.8x	0.8x	0.8x	0.8x	0.9x
Change in Revenue	-	(545)	1,118	823	494	703	686	705	853	1,035
Change in Total Asset	-	(123)	367	1,025	1,464	1,136	1,141	999	642	638
Reinvestment Ratio	-	4.4x	3.0x	0.8x	0.3x	0.6x	0.6x	0.7x	1.3x	1.6x
COGS	745	677	910	1,001	1,074	1,201	1,410	1,565	1,832	2,070
Inventory	60	48	59	72	110	99	116	120	141	159
Asset Turnover Ratio	3.0x	3.0x	0.8x	0.3x	0.7x	0.9x	0.7x	0.8x	1.3x	13.0x

Relative valuation - Benchmarking

Rationale for Selecting Comparable Companies

Apollo Hospital, Max Healthcare, Fortis Healthcare, KIMS, and Aster DM Healthcare are good comparable companies for Narayana Hrudayalaya because they operate in the hospital services sector and offer similar types of healthcare, including cardiac care, oncology, and orthopedics. These companies have a large network of hospitals and clinics across India, with a mix of hospitals they own and others they manage, just like Narayana Hrudayalaya.

This makes them useful for comparing key metrics like profitability, efficiency, use of medical technology, healthcare delivery, and growth. Smaller companies like Rainbow and Global Health don't fit as well for comparison because they operate on a smaller scale and have lower debt levels, making them less relevant for comparing industry-wide metrics.

Summary						
Benchmarking - Broad set	2020	2021	2022	2023	2024	Historical Average
Average Revenue Growth	14.08%	2.67%	40.07%	6.64%	15.99%	15.89%
Average EBITDA Margins	17.50%	16.78%	22.47%	22.36%	22.60%	20.34%
Average EBIT Margins	10.58%	9.35%	16.73%	16.80%	16.99%	14.09%
Average PBT Margins	7.05%	4.94%	16.13%	17.56%	16.05%	12.35%
Average PAT Margins	4.98%	2.99%	12.39%	14.58%	12.06%	9.40%
Median Revenue Growth	11.44%	-3.58%	41.87%	15.90%	14.73%	14.73%
Median EBITDA Margins	15.35%	13.61%	21.14%	23.06%	24.40%	21.14%
Median EBIT Margins	9.00%	5.64%	15.15%	17.50%	19.12%	15.15%
Median PBT Margins	5.17%	2.24%	17.28%	17.08%	18.40%	17.08%
Median PAT Margins	3.84%	1.99%	13.82%	15.88%	13.45%	13.45%
Benchmarking - Considered set	2020	2021	2022	2023	2024	Historical Average
Average Revenue Growth	12.59%	6.39%	36.18%	0.33%	15.98%	14.30%
Average EBITDA Margins	16.36%	15.68%	20.91%	19.88%	20.09%	18.59%
Average EBIT Margins	10.14%	9.25%	15.77%	14.74%	15.02%	12.99%
Average PBT Margins	6.42%	4.76%	16.14%	16.45%	14.10%	11.57%
Average PAT Margins	4.95%	2.57%	12.69%	14.37%	10.60%	9.04%
Median Revenue Growth	11.44%	-0.50%	38.85%	13.30%	14.73%	13.30%
Median EBITDA Margins	15.35%	12.81%	18.70%	17.49%	18.39%	17.49%
Median EBIT Margins	9.00%	5.64%	13.43%	12.47%	13.42%	12.47%
Median PBT Margins	5.17%	2.09%	17.28%	17.08%	12.45%	12.45%
Median PAT Margins	3.84%	1.30%	13.82%	15.88%	9.36%	9.36%

Relative valuation



In INR Crores

Market Data						Financials									Multiple (x)								
Relative Valuation	Ticker	Share Price	Market Cap	Debt	Enter price Value	2024A Sales	2025E Sales	2026E Sales	2024A EBITDA	2025A EBITDA	2026A EBITDA	2024A PAT	2025A PAT	2026A PAT	2024A EV/Sales	2025A EV/Sales	2026A EV/Sales	2024A EV/EBITDA	2025A EV/EBITDA	2026A EV/EBITDA	2024A P/E	2025A P/E	2026A P/E
Apollo Hospitals	APOLLOHOSP	6,924	99,856	5,333	1,04,255	19,059	22,285	26,061	2,391	3,164	3,956	935	1,487	2,046	5.5x	4.7x	4.0x	43.6x	33.0x	26.4x	106.8x	67.2x	48.8x
Max Healthcare	MAXHEALTH	938	91,186	1,299	91,386	5,406	8,021	10,125	1,492	2,261	2,875	1,058	1,536	2,014	16.9x	11.4x	9.0x	61.3x	40.4x	31.8x	86.2x	59.4x	45.3x
Fortis Healthcare	FORTIS	583	43,999	1,155	44,556	6,893	7,879	8,939	1,268	1,580	1,893	645	817	1,034	6.5x	5.7x	5.0x	35.1x	28.2x	23.5x	68.2x	53.9x	42.6x
KIMS	KIMS	523	20,917	1,355	22,224	2,498	2,953	3,556	640	781	916	336	373	443	8.9x	7.5x	6.2x	34.7x	28.5x	24.3x	62.3x	56.1x	47.2x
Global Health	MEDANTA	1,060	28,470	802	28,097	3,275	3,835	4,575	799	1,008	1,219	478	614	762	8.6x	7.3x	6.1x	35.2x	27.9x	23.0x	59.6x	46.4x	37.4x
Rainbow Medicare	RAINBOW	1,421	14,428	765	15,183	1,297	1,574	1,860	429	502	608	218	265	327	11.7x	9.6x	8.2x	35.4x	30.2x	25.0x	66.2x	54.4x	44.1x
Aster DM Healthcare	ASTERDM	449	22,416	1,940	22,779	3,699	4,307	4,979	578	745	919	212	351	493	6.2x	5.3x	4.6x	39.4x	30.6x	24.8x	105.7x	63.9x	45.5x

High		16.9x	11.4x	9.0x	61.3x	40.4x	31.8x	106.8x	67.2x	48.8x
75th Percentile		10.3x	8.6x	7.2x	41.5x	31.8x	25.7x	96.0x	61.6x	46.3x
Average		9.2x	7.4x	6.2x	40.7x	31.2x	25.5x	79.3x	57.3x	44.4x
Median		8.6x	7.3x	6.1x	35.4x	30.2x	24.8x	68.2x	56.1x	45.3x
25th Percentile		6.3x	5.5x	4.8x	35.2x	28.3x	23.9x	64.2x	54.1x	43.3x
Low		5.5x	4.7x	4.0x	34.7x	27.9x	23.0x	59.6x	46.4x	37.4x

Relative Valuation – EV/EBITDA	
Sales	5,018
EBITDA	1168
PAT	790
Valuation Multiple	EV/EBITDA
EV/EBITDA	35.4x
Enterprice value	41337
Less:Debt	1,220
Add:Cash	417
Less:Minority Interest	-
Add:Value of Holdings	-
Add:Non Operating Asset	4
Equity Value	42,978
Less: Value of Options	-
Equity Value for Shareholder	42,978

Relative Valuation – EV/Sales	
Sales	5,018
EBITDA	1168
PAT	790
Valuation Multiple	EV/Sales
EV/Sales	8.6x
Enterprice value	43051
Less:Debt	1,220
Add: Cash	417
Less:Minority Interest	-
Add:Value of Holdings	-
Add:Non Operating Asset	4
Equity Value	44,692
Less: Value of Options	-
Equity Value for Shareholder	44,692

Relative Valuation – P/E	
Sales	5,018
EBITDA	1168
PAT	790
Valuation Multiple	P/E
P/E	68.2x
Equity Value	53890
Less:Debt	-
Add:Cash	-
Less: Minority Interest	-
Add: Value of Holdings	-
Add:Non Operating Asset	-
Equity Value	53,890
Less: Value of Options	-
Equity Value for Shareholder	53,890

Comparable Company Discription

Company Name	Discription
Apollo Hospitals	Apollo Hospitals Enterprise Limited (AHEL) is a leading private sector healthcare provider in Asia. It has a robust presence across the healthcare ecosystem, including Hospitals, Pharmacies, Primary Care & Diagnostic Clinics. AHEL's Hospitals are situated in Chennai, Hyderabad, Delhi, Ahmedabad, Pune, Changanur, Coimbatore, Jaipur, Madurai, Anantapur, Nellore, Kurnool, Bhopal, Ranchi, Bilaspur and Bachel. Established in 1979, the company operates 73 hospitals with over 10,000 beds and a significant presence in outpatient care, diagnostics, and retail health. Key product offerings include outpatient clinics, diagnostics, day surgery centres, and pharmacy distribution through the Apollo HealthCo platform, which has established a large network of pharmacies and digital health services.
Max Healthcare	Max Healthcare Institute provider, engaged in provision of healthcare services through primary care dinics, multi-specialty hospitals/medical centers and super-specialty hospitals facilities. These include managed facilities' and medical facilities of third party healthcare providers with whom, the Company has entered into long term service contracts for providing operation and management, medical services, clinical, radiology, pathology and related healthcare services. It operates 20 healthcare facilities with around 4,300 beds across North India and includes state-of-the-art tertiary and quaternary care hospitals, secondary care hospitals, and medical centres. Max Healthcare's business segments include its core hospital operations, which provide tertiary and quaternary healthcare, along with Max@Home for home health services and Max Labs for diagnostic services. The company has a significant presence in North India, offering specialized care in various fields and also engaging in public-private partnerships for some of its facilities.
Fortis Healthcare	Fortis Healthcare Limited (FHL) is one of the largest healthcare services providers in India with 27 hospitals, ~4,500 operational beds and over 426 diagnostics centers as of 31 March, 2023. The Company offers a full spectrum of integrated healthcare services ranging from clinics to quaternary care facilities and a wide range of ancillary services to patients in areas such as cardiac care, orthopedics, neurosciences, oncology, renal care, gastroenterology and mother and child care. They are delivering quality healthcare services to patients in modern facilities using advanced technology. It also has presence in Dubai, Mauritius and Sri Lanka.
Krishna Institute of Medical Sciences	The Company is one of the largest corporate groups in India, with hospitals in Telangana, Andhra Pradesh & Maharashtra providing multi-disciplinary integrated services with a focus on tertiary and quaternary healthcare at affordable cost. It operate 12 multispecialty hospitals under the "KIMS Hospitals" brand, with an aggregate capacity of over 4,000 beds. It offer a comprehensive range of healthcare services in over 40 specialties and super specialties, including cardiac sciences, oncology, neurosciences, gastric sciences, orthopedics, organ transplantation, renal sciences and mother & child care.
Aster DM Healthcare	Aster DM offer a diverse portfolio of specialized brands that serve patients in Primary, Secondary, Tertiary, and Quaternary Care, Healthcare Retailing, Diagnostic Laboratories, Digital Health, and Medical Education. With its patient-centric approach to medical care, the Company offer one of the largest and most widespread networks of Hospitals and Clinics, with the most premium healthcare facilities that aspire to make a difference. Company is recognized as one of the largest private healthcare providers operating in multiple GCC states and an emerging player in India through its network of 32 hospitals, 127 clinics, 521 pharmacies, 205 labs & patient experience centers (PECs). The Company operates 17 hospitals in India with an installed bed capacity of 4317 beds, as of March, 2023.
Rainbow Children's Medicare	The Company is a leading chain of multi-specialty pediatric, and obstetrics & gynecology hospitals in India, operating 16 hospitals and three clinics in six cities, with a total bed capacity of 1,655 beds, five of which are NABH-accredited and three of which have 183 received EDGE certification. The core specialties are Pediatrics and Obstetrics & Gynecology and emergency services. Its Pediatric Services include newborn and pediatric intensive care, pediatric multi-specialty services, pediatric Quaternary care (including multi-organ transplants); whereas, Obstetrics & Gynecology services include normal and complex obstetric care, multi-disciplinary fetal care, perinatal genetic and fertility treatment. Rainbow has been pivotal in saving thousands of children. Its inpatient specialties for child care includes Neonatology, Pediatric Intensive Care, Neonatal Intensive Care, Neurology, Gastroenterology, Nephrology, Hematology and Oncology, Pediatric Surgery, Cardiology and Nutrition.
Global Health	Medanta was established to innovate healthcare in India, focusing on both scientific and empathetic aspects of medicine. It operates a network of five high-quality hospitals across North India, aiming to deliver patient-centric care with a strong emphasis on collaboration, research, and innovation. Global Health Limited, operating under the brand Medanta, provides multi-specialty tertiary care focused on various medical specialties including cardiac sciences, neurosurgery, oncology, and orthopedics. As of June 30, 2024, it has 2,890 operational beds across five cities and is expanding through new facilities, enhancing its services to meet growing patient demands, and incorporating state-of-the-art technology and infrastructure.
Poly Medicare	The Company is a manufacturer/producer of Medical Devices. It manufactures and supply, in India and internationally, a diverse portfolio of medical devices in the product verticals of infusion therapy, oncology, anesthesia and respiratory care, urology, gastroenterology, blood management and blood collection, surgery and wound drainage, dialysis, central venous access catheters, veterinary medical devices and others. Poly Medicare Limited is a prominent player in the medical devices space with a robust presence across 4 countries and a network of 12 state-of-the-art manufacturing plants. The company emphasizes innovation and operational excellence in its offerings.

DuPont Analysis

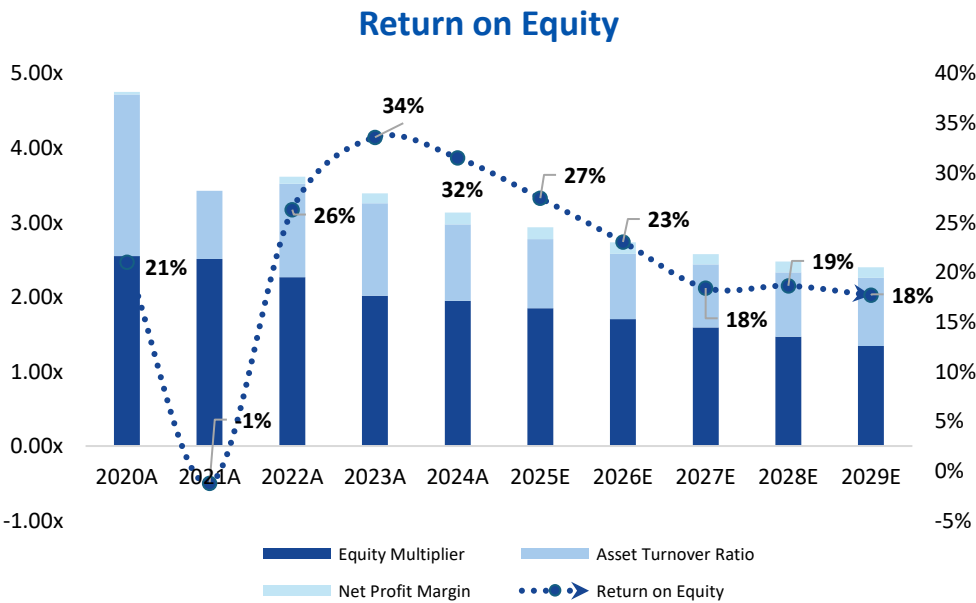
in INR Crores unless stated otherwise

						Explicit Forecast Period				
	FY2020	FY2021	FY2022	FY2023	FY2024	FY2025	FY2026	FY2027	FY2028	FY2029
Return on Equity (ROE)										
Net Profit	119	-14	342	607	790	917	990	971	1172	1296
Average shareholder's Equity	566	1124	1301	1808	2507	3342	4295	5275	6288	7334
Return on Equity	21%	-1%	26%	34%	32%	27%	23%	18%	19%	18%
ROE - Dupont Equation										
Net Profit	119	-14	342	607	790	917	990	971	1172	1296
Revnue	3128	2583	3701	4524	5018	5721	6407	7112	7965	9001
Net Profit Margin (A)	4%	-1%	9%	13%	16%	16%	15%	14%	15%	14%
Revnue	3128	2583	3701	4524	5018	5721	6407	7112	7965	9001
Average Total Assets	1443	2825	2947	3643	4887	6187	7325	8395	9216	9856
Asset Turnover Ratio (B)	2.17x	0.91x	1.26x	1.24x	1.03x	0.92x	0.87x	0.85x	0.86x	0.91x
Average Total Assets	1443	2825	2947	3643	4887	6187	7325	8395	9216	9856
Average shareholder's Equity	566	1124	1301	1808	2507	3342	4295	5275	6288	7334
Equity Multiplier (C)	2.55x	2.51x	2.27x	2.02x	1.95x	1.85x	1.71x	1.59x	1.47x	1.34x
Return on Equity (A*B*C)	21%	-1%	26%	34%	32%	27%	23%	18%	19%	18%
Return on Assets (ROA)										
Net Profit	119	-14	342	607	790	917	990	971	1172	1296
Average Total Assets	1443	2825	2947	3643	4887	6187	7325	8395	9216	9856
Return on Equity	8%	0%	12%	17%	16%	15%	14%	12%	13%	13%
ROA - Dupont Equation										
Net Profit	119	-14	342	607	790	917	990	971	1172	1296
Revnue	3128	2583	3701	4524	5018	5721	6407	7112	7965	9001
Net Profit Margin (A)	4%	-1%	9%	13%	16%	16%	15%	14%	15%	14%
Revnue	3128	2583	3701	4524	5018	5721	6407	7112	7965	9001
Average Total Assets	1443	2825	2947	3643	4887	6187	7325	8395	9216	9856
Asset Turnover Ratio (B)	2.17x	0.91x	1.26x	1.24x	1.03x	0.92x	0.87x	0.85x	0.86x	0.91x
Return on Assets (A*B)	8%	0%	12%	17%	16%	15%	14%	12%	13%	13%

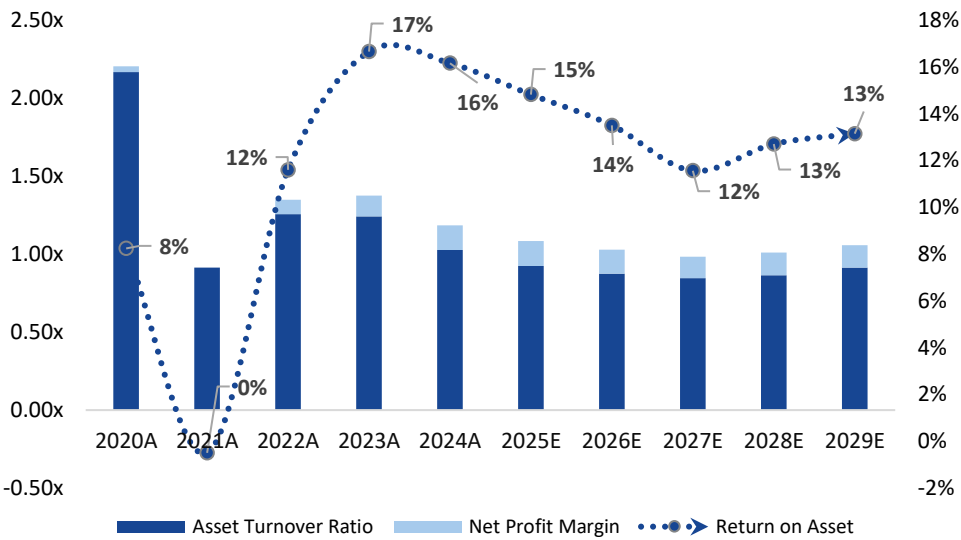
Return on Equity

Return on Equity (ROE) shows how well a company is using its shareholders' capital to make profits. We can break down ROE into three main parts: Net Profit Margin, Asset Turnover Ratio, and Equity Multiplier.

In our projections, the company's net profit margin and asset turnover ratio are expected to stay steady, showing that the company's operations remain consistent. However, ROE is projected to rise to around 32% in 2024 before gradually declining to 18% by 2029. This decline is largely due to the company's plan to reduce its reliance on debt, moving towards a safer and more balanced capital structure. As the company matures, the equity multiplier is expected to fall from 1.95x to 1.34x, which means the company will rely less on borrowed money. This shift will likely lower ROE from around 30%-34% down to 18%-23% over time. Even though this results in a smaller ROE, the company will be more financially stable and resilient, with a focus on steady, sustainable growth.



Return on Asset



Return on Asset

Return on Asset can be dissected into two key components: net margins and the asset turnover ratio. Variations in either of these factors will directly influence the ROA.

Over the past 2-3 years, the company's Return on Assets (ROA) has increased by about 4-5% due to an improvement in net profit margins, which have gone up by 5-6%. During this period, the asset turnover ratio has been relatively stable, ranging from 1.03x to 1.26x. As a result, the company's ROA has consistently stayed within a healthy range of 12% to 16%. This stability in ROA indicates that the company is effectively using its assets to generate returns, with improved profitability supporting this consistent performance.

Altman's Z-Score Analysis

in INR Crores unless stated otherwise

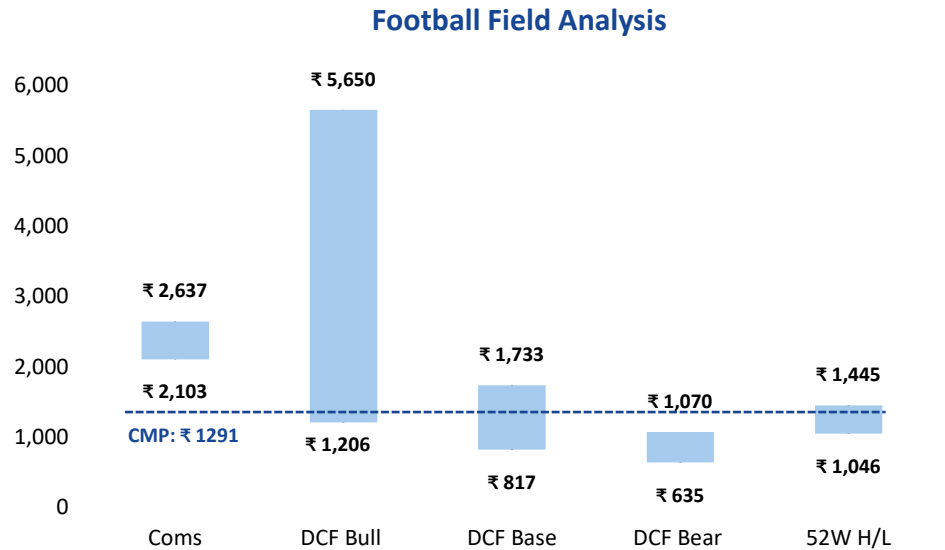
	Actuals				
	FY2020	FY2021	FY2022	FY2023	FY2024
Working Capital/Total Assets					
Working Capital	(20)	(40)	129	303	834
Total Asset	2886	2763	3130	4155	5619
Working Capital/Total Assets	-1%	-1%	4%	7%	15%
Retained Earnings/Total Assets					
Retained Earnings	99	(14)	321	558	711
Total Asset	2886	2763	3130	4155	5619
Retained Earnings/Total Assets	3%	-1%	10%	13%	13%
EBIT/Total Assets					
EBIT	240	7	480	769	926
Total Asset	2886	2763	3130	4155	5619
EBIT/Total Assets	8%	0%	15%	19%	16%
Market Cap/Long-Term Liabilities					
Market Cap	5059	8296	15235	15824	26241
Long-Term Liabilities	2875	2742	3119	4142	5594
Market Cap/Long-Term Liabilities	1.76x	3.03x	4.88x	3.82x	4.69x
Revenue/Total Asset					
Revenue	3128	2583	3701	4524	5018
Total Asset	2886	2763	3130	4155	5619
Revenue/Total Asset	1.08x	0.93x	1.18x	1.09x	0.89x
Altman's Z Score					
Z Score	2	3	5	4	5
Financial Stability	Grey Zone	Grey Zone	Strong	Strong	Strong

- Altman's Z-Score is a metric which takes into account different aspects of a company's financials such as its retained earnings, EBIT, equity market value, sales, etc. in particular weights. The final output tells us about a company's solvency strength and if the company is likely to go bankrupt over the next two years.
- Z - score above 3 signifies that the company is financially stable and is not expected to face solvency issues in near future. market cap of the company and revenue to total assets is constantly increasing, thereby boosting the Z-Score over the years.

DCF Value

According to the DCF valuation approach, the intrinsic value of Narayana hrudaya shares varies significantly based on the discount rate applied. Under a bearish outlook, with a high discount rate of approximately 12%, the intrinsic value ranges between ₹635 and ₹1,070. In a neutral scenario, using a moderate discount rate of around 10%, the value lies between ₹817 and ₹1,733.

For a bullish outlook, with a low discount rate of roughly 8%, the share value is estimated to be between ₹1,206 and ₹5,650. These valuations are sensitive to the growth rate assumptions during both the explicit forecast period and the terminal period. With the current market price hovering around ₹1,291 as of November 2024, the analysis suggests that the company's shares are undervalued in a Base and Bull DCF scenarios.



Relative Value

According to the Relative Valuation approach, the intrinsic value of Narayana hrudaya shares is estimated at ₹2,186 based on the EV/Revenue multiple (~8.6x), ₹2,103 using the EV/EBITDA multiple (~35.4x), and ₹2,637 based on the P/E multiple (~68.2x).

These multiples were derived as the median values from a group of seven companies in the healthcare sector, though they may fluctuate over time. With the current market price of the shares around ₹1,291 as of November 2024, the analysis suggests that the shares are undervalued for the EV/EBITDA, EV/Revenue and P/E multiples used in the relative valuation approach.

Method & Case	Equity Value		Share Price	
	High	Low	High	Low
DCF Bull	₹ 1,15,496	₹ 24,649	₹ 5,650	₹ 1,206
DCF Base	₹ 35,427	₹ 16,695	₹ 1,733	₹ 817
DCF Bear	₹ 21,863	₹ 12,975	₹ 1,070	₹ 635
Relative-EV/Revenue	₹ 44,692		₹ 2,186	
Relative-EV/EBITDA	₹ 42,978		₹ 2,103	
Relative-P/E	₹ 53,890		₹ 2,637	
No. of Shares	20.44Cr.			
Current Market Price	₹ 1291			

Valuation Method	Enterprise value	Equity value Available for Shareholders	Value per share	Market Price	Discount/Premium	Discount/ Premium%
Discounted Cashflow- Perpetuity Growth	18732	20372.9	997	1291	294	22.8%
Discounted Cashflow- Exit Multiple	58937	60578.3	2964	1291	(1,673)	-129.6%
Relative Valuation -EV/Sales	43051	44691.6	2186	1291	(895)	-69.4%
Relative Valuation -EV/EBITDA	41337	42978.4	2103	1291	(812)	-62.9%
Relative Valuation - P/E	-	53890.2	2637	1291	(1,346)	-104.2%

Analysis

According to DCF Valuation perpetuity growth method share is trading at a premium of INR 294 and exit multiple share trade discounted INR 1673 respectively. However, according to relative valuation the share is trading at a discounted of all the multiples EV/Sales, EV/EBITDA and P/E INR 895, 812 and 1346 respectively.

Analysis Considering DCF Valuation for Perpetual Growth

The DCF valuation using Perpetuity Growth shows that the share price is higher than its value, with a premium of INR 294, meaning it may be overvalued from a long-term perspective.

- ☐ **Market Share and Growth:** The company is focusing on both expanding on its own (organic growth) and buying other businesses (inorganic growth) in India and abroad. They aim to create benefits that fit well with their current operations and bring value to stakeholders. The company believes it can achieve strong, double-digit revenue growth in the coming years.
- ☐ **Operational Efficiency:** The company is working to improve clinical and non-clinical operations, upgrade patient services, increase patient numbers, build capacity, invest in digital channels for patients, and become more efficient overall.
- ☐ **Industry Demand:** Demand for healthcare services in India is expected to keep growing due to an aging population and overall population growth. There is "sufficient demand" for healthcare services, even if population numbers and age groups stay the same.
- ☐ **Perpetual Growth:** The DCF model assumes moderate, steady growth in the long run, balancing possible risks and opportunities.

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